A Symmetry-Quotient Reduction for Odd-Characteristic SNOVA: Exact Structure and Conditional Recovery Ledgers

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Abstract

After the attacker-imposed common-column restriction used in affine-column attacks, odd-characteristic ($q = 19$) SNOVA Version 2.3 loses quadratic dimension. If $A = \mathbb{F}_q[S]$ has dimension d, symmetry makes the restricted scalar features indexed by (ξ, η) and (η, ξ) identical, so the d^2 labels factor through $\mathrm{Sym}^2(A)$ and span at most $\binom{d+1}{2}$ dimensions: 16 becomes 10 for $\ell = 4$, and 4 becomes 3 for $\ell = 2$. This is not a rank bound on the unrestricted matrix-valued verifier. Conditional on exact public quotient/rank preflights, an output split yields square residual systems while retaining every verifier equation and the original rejection rule.

Under the explicitly inherited, unproved hypothesis $\mathsf{H}_{\mathrm{hom}}$, a finite-cardinality symbolic-homotopy adaptation gives conditional selected-degree-free recovery routes. Their worst adaptive exponents are 132.188, 183.795, and 233.844 in a unit-leading multiplication schedule, before an unmeasured multiplier κ_{hom}; they are break-even ledgers, not complete classical-gate bounds. A separate Level-I $\ell = 2$ Boolean sensitivity applies the published *Just Guess* transformation. Exact enumeration of the official local channel transform bounds every nonzero conditional cross-channel functional atom by 49/829. Under the published random-system premise, the per-trial ledger is below $2^{132.047}$ gates. If a trial supplies at least $19^{16}/4$ sign-canonical diagonal roots, a second moment gives success probability above 0.0961 and cost below $2^{135.425325}\kappa_{\mathrm{JG}}$. Candidate abundance, restarts, and excess processing remain inside the unmeasured κ_{JG}; a sufficient nominal crossing condition is $\log_2 \kappa_{\mathrm{JG}} < 7.574675$.

The release checks formulas and circuits, instantiates the public reduction on one official KAT, and produces a non-planted zero-offset forgery only at an unofficial toy shape. That test interpolates the complete restricted verifier and exercises its rank-three quotient, target filter, reconstruction, and rejection paths. It neither runs a production-size solver nor demonstrates an EUF-CMA forgery at a published set. The accompanying output-increase percentages are necessary dimension floors within the common-column model, not security sufficiency claims.

1 Introduction

NIST's Additional Digital Signatures process seeks signature schemes with security and performance profiles that complement the standards selected in its first post-quantum process. SNOVA is one of the nine candidates advanced to the third round in May 2026 [1, 2]. It belongs to the Unbalanced Oil and Vinegar line of multivariate signatures, where verification evaluates a public quadratic map and signing uses a hidden central map with an oil-and-vinegar structure [3, 4].

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SNOVA obtains compact public keys by representing variables and coefficients with small matrices and generating many public forms from a smaller structured family. That structure has driven its cryptanalysis. The first analyses scalarized or lifted the secret core to UOV systems and applied key-recovery attacks. Beullens instead exposed a structured MAYO-style outer map and obtained affine-column forgery attacks. Subsequent work improved both lines through group actions, extension-field lifting, and multihomogeneous solving [5–10]. These works analyze the original or Round 2 construction over $\mathbb{F}_{16}$.

The next revision was prompted by a different attack. Ran's wedge method uses the alternating polar forms available in characteristic two, and its SNOVA adaptation exploits the scheme's block-ring structure [11, 12]. In response, the SNOVA team proposed odd-prime fields together with symmetric public quadratic matrices at the sixth NIST PQC conference. The team also warned that symmetrization could introduce new attacks and that its effect on wedge-style cryptanalysis was not yet settled [13]. NIST later described this odd-characteristic symmetric construction as a response to the wedge attack and did not regard SNOVA as having reached a stable form [2].

This change entered the specification in Version 2.1, not Version 2.3. Version 2.3 subsequently added rectangular signatures and revised parameter sets [4]. We use "Version 2.3" because it is the current specification and because our results cover its nine published $q = 19$ parameter shapes. The algebraic design choice under study is the odd-characteristic symmetrization inherited from Version 2.1.

Odd characteristic alone is not an unexplored cryptanalytic regime. Jin et al. generalized the exterior-algebra attack to arbitrary characteristic by introducing a reduced symmetric algebra [14]. Their attack is a generic UOV key-recovery method, and their concrete odd-characteristic study concerns QR-UOV. It neither analyzes the symmetric public matrices of the current SNOVA specification nor composes their effect with an affine-column forgery relation.

The question is therefore what the public symmetrization does after the common-column restriction used in affine-column attacks. Under this public restriction, the Version 2.3 verifier factors through a symmetric-square quotient. An exact public output split then converts its equations into square residual systems for every published $q = 19$ parameter shape that passes the stated public rank preflights.

This statement concerns the restricted verifier, not the unrestricted matrix-valued map. Let $A = \mathbb{F}_q[S]$ have dimension d, and let $\Sigma_\xi = I_n \otimes \xi$ for $\xi \in A$. After all signature columns are expressed through one stacked column u, symmetry gives $q_{i,\xi,\eta}(u) = q_{i,\eta,\xi}(u)$. The d^2 ordered labels therefore factor through $\mathrm{Sym}^2(A)$, with the exact dimension change

$$d^2 \longrightarrow \binom{d+1}{2}, \qquad 16 \longrightarrow 10 \text{ for } d = 4, \qquad 4 \longrightarrow 3 \text{ for } d = 2. \tag{1}$$

The full verifier relates the two ordered features by matrix transposition; only the common-column restriction makes them the same scalar polynomial. The identification summarized in Equation (1) is exact and basis-independent, but deliberately local to the attacker-imposed restriction (Theorem 3.1).

1.1 Our results

An exact symmetry quotient. We prove that every symmetric public base form factors through $\mathrm{Sym}^2(A)$ after the common-column restriction. The proof occurs before the outer public mixing, so later linear combinations cannot recover the alternating directions that have vanished. For the published choices $d = \ell$, the resulting caps are exactly those in (1) (Theorem 3.1 and table 3).

A verifier-preserving square reduction. We compose the quotient with the affine-column relation of Beullens [7]. Public row reduction separates output coordinates reached by the quotient quadratics from affine equations and target-consistency checks. The remaining nonlinear coordinates form a square residual system on a public slice. Reconstruction, rejection, and the final unmodified verifier check remain part of the wrapper (Theorems 4.2, 5.4 and 12.1).

Conditional recovery ledgers. We give a selected-degree-free Las Vegas recovery theorem for all published parameter shapes and an adaptive version that uses exact public determinant tests to select smaller cores. The corresponding worst levelwise arithmetic exponents are 142.607, 185.451, and 233.844 for the uniform square route, and 132.188, 183.795, and 233.844 for the adaptive route (Theorems 12.1 and 12.2). These values assume the explicit hypothesis H_{hom} and omit the unmeasured multiplier κ_{hom}. A separate Level-I sensitivity instantiates the published Just Guess schedule but retains its own unmeasured κ_{JG} (Theorem 10.3). The ledgers are conditional break-even calculations, not complete classical-gate breaks.

Evidence and design implications. The artifact checks the algebra and ledgers, instantiates the public reduction on one official known-answer-test key, and exercises the full forgery wrapper only at an unofficial toy shape. It contains no production-size solve or forgery (Table 4). We also derive necessary output floors for repairs that retain the current common-column architecture. The combined floor requires increases between 45.00% and 60.71%, while the narrower zero-offset condition requires increases between 101.5625% and 144.00% (Theorems 15.2 and 15.4). These are necessary dimension conditions for the stated repair goals, not sufficient conditions for security.

The claim hierarchy is intentional. The quotient, output split, reconstruction, and verifier check are exact public computations once their displayed rank checks pass. The fixed-layout and realized-rank conditions are deterministic preflights, but this paper does not estimate their pass rate over official keys. Root-abundance statements use either an idealized transcript model or a fixed-key certificate. Recovery costs additionally depend on H_{hom} and the unmeasured implementation multipliers. Nothing in the paper establishes an unconditional production-size forgery.

1.2 Technical overview

The attack begins with the affine-column relation introduced in earlier forgery attacks. For each matrix-valued signature block, the attacker writes all columns as public affine functions of one stacked column u. Substituting this relation into the public verifier leaves $N_0 = n\ell$ scalar variables and turns every ordered power-pair feature into a scalar quadratic polynomial $q_{i,\xi,\eta}(u)$. This restriction is chosen by the attacker and does not assert an identity on the unrestricted matrix-valued map.

The new observation occurs before the outer public mixing. Because the Version 2.3 base matrices are symmetric, the restricted scalar polynomials satisfy $q_{i,\xi,\eta} = q_{i,\eta,\xi}$. The ordered $A \otimes A$ feature space therefore factors through $\mathrm{Sym}^2(A)$. For $\ell = 4$, each group of sixteen ordered labels contributes at most ten distinct quadratics; for $\ell = 2$, four labels contribute at most three. Later public linear maps cannot recover the alternating directions removed by this quotient.

For a fixed public column relation R, the attack computes the linear map Q_R from quotient features to verifier outputs. A left inverse G_R extracts the coordinates reached by the quotient quadratics, while a basis W_R of the left kernel extracts the remaining affine constraints and target consistency conditions. This is an invertible public change of output coordinates. It does not discard equations or replace the verifier with a surrogate.

The affine constraints define a public solution space. The attack chooses a slice L whose dimension equals the number of surviving quotient quadratics, so the nonlinear part becomes a square residual system. The structural preflight checks the ranks needed for this split and for the fresh coefficient blocks used by the slice. The target filter and rejection analysis separately ensure that a recovered point corresponds to the exact official target distribution and satisfies the signature-format rules.

What remains is a root-recovery problem, and this is where the conditional analysis begins. The spectrum bounds state conditions under which a slice contains an accepted nonsingular root. Under H_{hom}, a finite-cardinality symbolic-homotopy algorithm recovers such roots with the reported arithmetic ledgers. Exact determinant tests may select smaller cores, but the complete square system remains the fallback. These steps do not turn the conditional solver estimate into an unconditional break.

Finally, every recovered root is lifted from the slice, reconstructed into a signature candidate, subjected to the public rejection rules, and checked by the unmodified verifier. A failed target filter, preflight, solve, or final verification causes a restart. Thus the quotient, output split, reconstruction, and verifier check are exact public computations; root abundance and production-size recovery costs retain the explicit conditions stated in the theorems.

Roadmap. Section 2 gives the MQ background and situates the result among earlier SNOVA attacks. Sections 3 and 4 prove the symmetry quotient and exact output split. Section 5 handles public slices, targets, and rejection. The root and square-recovery theorems begin in Section 6; the Just Guess sensitivity appears in Section 10. The remaining sections give the low-state XL complement, artifact evidence, repair bounds, and conclusions.

2 Background and relation to prior work

2.1 MQ systems, roots, and affine slices

An MQ system over $\mathbb{F}_q$ is a map $g : \mathbb{F}_q^N \to \mathbb{F}_q^K$ whose coordinates have total degree at most two, together with a target $z \in \mathbb{F}_q^K$. Linear and constant terms are allowed. A root is an x satisfying $g(x) = z$. Underdetermination makes roots more plentiful but does not make the system linear or automatically easy.

An affine space is a translate $a + V$ of a linear subspace. Restricting the search to an affine slice $a + L \subseteq a + V$ turns the problem into a system in $h = \dim L$ coordinates. Three events must be separated:

(i) the target–slice pair contains a base-field root;
(ii) it contains an accepted root at which a chosen Jacobian has full rank;
(iii) the chosen algebraic solver recovers every such root within its stated work bound.

The first two are properties of the map and slice. The third is a solver statement. The paper proves them separately.

2.2 Polar forms, rank spectra, and collisions

For a scalar quadratic polynomial f, define the polar form

$$\beta_f(x, y) = f(x + y) - f(x) - f(y) + f(0).$$

In odd characteristic this is bilinear. For a vector-valued map g, a nonzero $b \in \mathbb{F}_q^K$ selects the scalar quadratic $b \cdot g$. Its polar rank measures how many independent directions the quadratic part couples.

For a fixed slice direction y, the vector-valued derivative

$$T_y(x) = g(x+y) - g(x) - g(y) + g(0)$$

is linear in x. The value $q^{-\operatorname{rank} T_y}$ is exactly the fraction of its codomain-dual directions that annihilate T_y, after the appropriate normalization. Summing these quantities over nonzero y controls the second factorial moment of the number of roots in a random target–coset pair. The same spectrum controls the density of points where the restricted Jacobian loses rank.

2.3 Macaulay matrices and the low-state branch

XL multiplies the input quadratics by selected monomials and treats the resulting polynomials as rows of a Macaulay matrix. Generic XL tracks total degree. Multigraded XL tracks one degree per variable block and can be much smaller when the system has a block structure [8, 9]. A one-dimensional right kernel is the easiest extraction case, but it is not the only sound one. If row reduction supplies a commuting border basis, the resulting multiplication matrices describe a finite quotient algebra; FGLM and finite-field factorization then enumerate its roots [15, 16]. The streamed-XL section later makes the additional production-degree premise fully explicit.

2.4 Finite-cardinality symbolic homotopy

The selected-degree-free branch adapts a multihomogeneous homotopy algorithm. The scope of the adaptation is important.

Partition $N = \sum_{j=1}^{b} n_j$ variables into b blocks $X_1, \ldots, X_b$. Let $f_1, \ldots, f_N$ be nonconstant polynomials represented by a straight-line program of length L, and suppose f_i has multidegree at most $\mathbf{d}_i = (d_{i,1}, \ldots, d_{i,b})$. Define

$$B = C_{\mathbf{n}}(\mathbf{d}) = [\vartheta_1^{n_1} \cdots \vartheta_b^{n_b}] \prod_{i=1}^{N} \left(\sum_{j=1}^{b} d_{i,j} \vartheta_j \right), \tag{2}$$

$$B^+ = C'_{\mathbf{n}}(\mathbf{d}') = \operatorname{coeffsum} \left(\prod_{i=1}^{N} \left(\vartheta_0 + \sum_{j=1}^{b} d_{i,j} \vartheta_j \right) \bmod \langle \vartheta_0^2, \vartheta_1^{n_1+1}, \ldots, \vartheta_b^{n_b+1} \rangle \right). \tag{3}$$

Inherited homotopy hypothesis $\mathsf{H}_{\mathrm{hom}}$. For the input system and the chosen random parameters, assume that the incidence, deformation-curve regularity, specialization, and cleaning hypotheses used by Proposition 5, Algorithm 2, and Remark 15 of Safey El Din and Schost remain valid after the two finite-field substitutions made below [17]. In particular, the relevant homotopy components must have the expected dimensions and the specializations and inversions used during Newton lifting must avoid their exceptional loci. This paper proves the start-system and separator substitutions below; it does not derive $\mathsf{H}_{\mathrm{hom}}$ from the specific ideals produced by SNOVA, and no released script tests it at a published parameter set.

Theorem 2.1 (Conditional finite-cardinality homotopy adaptation)**.** *Assume* $\mathsf{H}_{\mathrm{hom}}$. *Let k be a perfect finite field and put*

$$E_0 = \max \left\{ \max_j \sum_i d_{i,j},\ 8B^2 \right\}.$$

If $|k| \geq E_0$, a randomized symbolic-homotopy algorithm whose separating form has a coefficient vector sampled uniformly from k^N returns a zero-dimensional parametrization containing every *nonsingular root with probability at least* $7/8$*. On every run it returns a subset of the nonsingular roots or reports failure. Its arithmetic cost is*

$$\widetilde{O}\left(BB^{+}\left(L+\sum_{i,j} d_{i,j}+N^{2}\right)N\right)$$

operations in k*. This soft-O theorem does not instantiate its implied constant or polylogarithmic factors. More generally, if* $|k| \geq \max_j \sum_i d_{i,j}$ *and* $|k| > B^2$*, the same algorithm succeeds with probability at least* $1 - B^2/|k|$*.*

Proof. Proposition 5 and Algorithm 2 of Safey El Din and Schost give the underlying homotopy construction under their stated hypotheses, with the bad-run behavior described in their Remark 15 [17]. The statement here is a new finite-cardinality adaptation, not a verbatim corollary of that proposition. Its two replacements for the cited characteristic-dependent choices are proved next.

For block j, put $D_j = \sum_i d_{i,j}$ and choose distinct $\alpha_{j,0}, \ldots, \alpha_{j,D_j-1} \in k$. The affine moment-curve forms

$$\kappa_{j,u}(X_j) = X_{j,1} + \alpha_{j,u} X_{j,2} + \cdots + \alpha_{j,u}^{n_j-1} X_{j,n_j} + \alpha_{j,u}^{n_j}$$

have the same Vandermonde properties as the start forms in the cited proof: any n_j are independent and any $n_j + 1$ are inconsistent. The product start system can be written explicitly. Put

$$s_{i,j} = \sum_{\nu=1}^{i} d_{\nu,j}, \qquad s_{0,j} = 0,$$

and, for $1 \leq i \leq N$, set

$$g_i(X) = \prod_{j=1}^{b} \prod_{u=s_{i-1,j}}^{s_{i,j}-1} \kappa_{j,u}(X_j).$$

The factor ranges are disjoint. An isolated common zero chooses, for each i, one vanishing factor of g_i; exactly n_j choices must come from block j. The coefficient in (2) counts these choices, including the $d_{i,j}$ possible factors. The Vandermonde property gives one point for each choice, while inconsistency of any $n_j + 1$ forms prevents additional vanishing factors. The corresponding block-diagonal linear Jacobian is nonsingular, so all B roots are simple. They are enumerated within the same soft-linear cost. The first term in E_0 supplies enough distinct nodes.

The homotopy proof next requires a well-separating linear form. It identifies at most B^2 nonzero vectors on which the coefficient vector must not vanish. The cited proof samples from a one-parameter moment-curve family, which is why its field-size condition includes an additional factor depending on N. Here the coefficient vector is instead sampled uniformly from the full space k^N. For each fixed nonzero vector, its annihilator inside k^N is a proper k-linear subspace (possibly of codimension greater than one), so a uniform coefficient vector annihilates it with probability at most $|k|^{-1}$. Since $|k| \geq 8B^2$, a union bound gives success at least $7/8$.

Under $\mathsf{H}_{\mathrm{hom}}$, the remaining Newton lifting, specialization, squarefree cleaning, and zero-dimensional parametrization steps of Algorithm 2 are algebraic over a perfect field and use only field operations and inversion at nonsingular points. Finite fields are perfect, so those correctness and arithmetic arguments carry over. Direct substitution into the original system and a Jacobian-rank test remove every spurious or singular output; hence a run returns only nonsingular roots or reports failure. As in the cited Remark 15, a bad run may return a strict subset, and completeness is not certified at comparable cost. The attack wrapper therefore treats absence of an accepted root as a restart rather than as a proof that none exists. □

The conditional theorem is a time theorem and does not promise small resident state. A parametrization can contain $O(B)$ field elements, but this output scale is dominated asymptotically by the displayed BB^+ term, which is typically $\Theta(B^2)$. All soft-O factors, fixed transforms, factorization, filtering, memory traffic, and implementation constants are retained later in the route- and parameter-dependent multiplier κ_{hom}.

2.5 SNOVA's attack and revision history

The first line of SNOVA cryptanalysis attacks the hidden UOV structure. Ikematsu and Akiyama analyze key recovery on the scalar expansion of the core map, and Li and Ding sharpen the interpretation as a UOV system with ℓv vinegar variables and ℓo oil variables over the base field [5, 6]. Nakamura, Tani, and Furue instead lift the same structure to a smaller UOV instance over $\mathbb{F}_{q^\ell}$ [10]. These attacks seek the hidden oil space or an equivalent signing key. They concern the original characteristic-two construction and do not use the symmetric public matrices introduced later.

Beullens exposes a different layer. He rewrites SNOVA as a structured MAYO-style whipping of a bilinear map, chooses an affine relation among the signature columns, and searches for rank-deficient combinations in the outer coefficient map [7]. The result is a forgery attack, including weak-key instances much cheaper than the generic case. The Round 2 changes vary the outer coefficient matrices and mitigate the largest rank drops, but they retain the affine-column attack surface. Our attack starts from this public restriction and acts one layer earlier: symmetry identifies the restricted power-pair polynomials before the outer map sees them.

Cabarcas et al. and Furue et al. improve the algebraic solving of the systems arising in the earlier attacks [8, 9]. The former exploits stability under a commutative group action and adapts the result to reconciliation and direct forgery. The latter transforms the systems over an extension field and uses their multihomogeneous structure. These works reduce the cost of solving an already identified SNOVA system. They do not derive an output-dimension loss from odd-characteristic public symmetrization.

Ran's wedge attack is a key-recovery attack on characteristic-two UOV. It uses the fact that polar forms are alternating and recovers the hidden oil space through exterior-algebra linearization [11]. Bros et al. specialize this method to SNOVA by exploiting its block-ring structure and give conjectural estimates below the claimed security levels for every Round 2 parameter set [12]. The SNOVA team's subsequent wedge-map analysis computes the relevant kernel dimensions for the Round 2 construction [18]. All three analyses concern the characteristic-two geometry that motivated the odd-prime revision.

Jin et al. give the main prior attack that genuinely crosses the characteristic boundary [14]. Their p-reduced symmetric algebra recovers Ran's exterior algebra at $p = 2$ and defines an XL key-recovery attack for arbitrary characteristic. Their concrete odd-characteristic application is to the twelve QR-UOV parameter sets, where the attack does not beat reconciliation on the recommended parameters. The method targets the hidden UOV oil space. It does not analyze the symmetric public-matrix encoding of odd-characteristic SNOVA, and it does not produce the verifier-preserving direct-preimage reduction used here.

The distinction between these attacks is summarized in Table 1. The exact new statement in this paper is the $\mathrm{Sym}^2(A)$ factorization after the public common-column restriction and the resulting output split. The direct forgery ledgers built on that reduction remain conditional on their stated root-spectrum and solver hypotheses.

Table 1: Scope of the closest attacks. "Output" is what a successful attack recovers; it is not a claim that every cited estimate is practical.

Work	Targeted regime	Main mechanism	Output and relation to this paper
Ikematsu–Akiyama; Li–Ding; Nakamura–Tani–Furue	Original / early $q = 16$	Scalar or extension-field UOV interpretation	Oil-space or equivalent-key recovery; no odd-q public symmetrization.
Beullens; Cabarcas et al.; Furue et al.	Original and Round 2 $q = 16$	Affine-column rank drop, group action, and multihomogeneous solving	Direct forgery or key recovery; supplies the public restriction that this paper composes with symmetry.
Ran; Bros et al.; Tseng et al.	Characteristic two, including Round 2 SNOVA	Exterior algebra and the wedge-map kernel	Oil-space recovery; motivates the odd-prime revision but does not apply the present quotient.
Jin et al.	UOV families in arbitrary characteristic; concrete odd-q QR-UOV instances	p-reduced symmetric-algebra XL	Oil-space recovery; no analysis of symmetric odd-q SNOVA public matrices.
This paper	All nine $q = 19$ Version 2.3 shapes	Common-column restriction, exact $\mathrm{Sym}^2(A)$ quotient, and public output split	Conditional direct preimages checked by the original verifier; no hidden-key recovery claim.

The complementary generic-MQ and partitioning literature includes Hashimoto's underdefined optimizer, Just Guess, generalized partitioning, and recent F4 engineering [19–22].

2.6 Notation at a glance

Table 2: Recurring notation.

Symbol	Meaning	Symbol	Meaning
(v, o, ℓ, r)	Vinegar blocks, oil blocks, block height, and columns per block.	$n = v + o$	Matrix-valued signature blocks.
$M = or\ell$	Scalar verifier outputs.	$N_0 = n\ell$	One stacked column after the common-column relation.
$A = \mathbb{F}_q[S]$	Public commutative matrix algebra; here $A \cong \mathbb{F}_{q^\ell}$.	$d = \dim_{\mathbb{F}_q} A$	Algebra degree; $d = \ell$ in every official row.
$m_1 = \lceil or/\ell \rceil$	Public base-form groups.	$K = m_1 \binom{\ell+1}{2}$	Maximum distinct quotient quadratics.
Q_R	Quotient-feature-to-output map for relation R.	W_R, G_R	Left-kernel basis and left inverse for the output split.
$C_{R,v}$	Affine constraints after projecting away the quadratic image.	L, s, h	Solver slice, its A-dimension when applicable, and base-field dimension.
$\bar{\eta}_L^U, \eta_{128}$	Directional collision mass and instantiated allowance.	$\rho_{\mathrm{byte}} = 14/256$	Maximum atom of an XOF byte reduced modulo 19.
$\rho_{\mathrm{cond}} = 49/829$	Conditional atom used by the Level-I cross-channel filter.	AXN	Two-input AND/XOR/XNOR ledger with free wiring and fan-out.
κ_{hom}	Complete homotopy work divided by its unit-leading ledger.	κ_{JG}	Multiplier for candidate regularity, restarts, and work outside the Just Guess ledger.

3 Public forms and the symmetry quotient

This section records the scalar quotient used after the common-column relation in Section 4. It is placed first to isolate the algebraic identity. Before that restriction, the verifier feature $U^{\mathsf{T}}\Sigma_\xi P_i \Sigma_\eta U$ is $r \times r$ and swapping (ξ, η) transposes this matrix; no entrywise equality is asserted.

3.1 Parameters and public forms

The public parameter tuple is (v, o, ℓ, r). There are $n = v + o$ matrix-valued signature blocks, each with ℓ rows and r columns. Stacking one column of all blocks gives a vector of length $N_0 = n\ell$. The verifier has

$$m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \qquad M = or\ell, \qquad N_0 = n\ell. \tag{4}$$

The ceiling is the convention used by the pinned Version 2.3 reference implementation [4].

Let $S \in \mathrm{Mat}_\ell(\mathbb{F}_q)$ be the public symmetric matrix with irreducible characteristic polynomial. Its powers generate

$$A = \mathbb{F}_q[S], \qquad \Sigma_\xi = I_n \otimes \xi \quad (\xi \in A).$$

For the public rows, $A \cong \mathbb{F}_{q^\ell}$ and $\dim_{\mathbb{F}_q} A = \ell$. The scalar-expanded base matrices $P_1, \dots, P_{m_1} \in \mathrm{Mat}_{N_0}(\mathbb{F}_q)$ are symmetric. For $\xi, \eta \in A$, define

$$B_i^{\xi,\eta}(x, y) = x^{\mathsf{T}}\Sigma_\xi P_i \Sigma_\eta y. \tag{5}$$

Setting $x = y$ gives the quadratic power-pair feature.

Table 3: Dimensions of the nine public $q = 19$ Version 2.3 parameter rows.

Level	(v, o, ℓ, r)	m_1	M	K	N_0
I	$(28, 5, 4, 4)$	5	80	50	132
I	$(48, 16, 2, 2)$	16	64	48	128
I	$(28, 4, 4, 5)$	5	80	50	128
III	$(40, 7, 4, 4)$	7	112	70	188
III	$(72, 24, 2, 2)$	24	96	72	192
III	$(38, 5, 4, 5)$	7	100	70	172
V	$(50, 9, 4, 4)$	9	144	90	236
V	$(96, 32, 2, 2)$	32	128	96	256
V	$(52, 6, 4, 6)$	9	144	90	232

3.2 The basis-independent quotient

For one base form in the restricted scalar map, the ordered label space is $A \otimes_{\mathbb{F}_q} A$. The following theorem identifies its universal kernel.

Theorem 3.1 (Symmetric-square factorization)**.** *Let $A = \mathbb{F}_q[S]$ have dimension d, and assume $S^{\mathsf{T}} = S$ and $P_i^{\mathsf{T}} = P_i$. For each i, the map*

$$A \otimes_{\mathbb{F}_q} A \longrightarrow \mathbb{F}_q[u]_2^{\mathrm{hom}}, \qquad \xi \otimes \eta \longmapsto u^{\mathsf{T}}\Sigma_\xi P_i \Sigma_\eta u$$

is invariant under $\xi \otimes \eta \mapsto \eta \otimes \xi$. It factors through

$$\mathrm{Sym}^2_{\mathbb{F}_q}(A) = (A \otimes_{\mathbb{F}_q} A)/\operatorname{span}\{\xi \otimes \eta - \eta \otimes \xi\}.$$

Consequently the direct sum over m_1 base forms has rank at most $m_1\binom{d+1}{2}$. After any public linear map to M outputs, its rank is at most

$$\min\left\{M,\ m_1\binom{d+1}{2}\right\}. \tag{6}$$

If the minimal polynomial of S has degree ℓ, then $d=\ell$ and the ordered feature vector is obtained by duplicating the off-diagonal coordinates of an unordered feature vector.

Proof. Every element of A is symmetric. Since the displayed value is a scalar,

$$u^{\mathsf{T}}\Sigma_\xi P_i \Sigma_\eta u = \left(u^{\mathsf{T}}\Sigma_\xi P_i \Sigma_\eta u\right)^{\mathsf{T}} = u^{\mathsf{T}}\Sigma_\eta P_i \Sigma_\xi u.$$

Thus every alternating label $\xi\otimes\eta-\eta\otimes\xi$ lies in the kernel. The symmetric square has dimension $\binom{d+1}{2}$, and a later linear map cannot enlarge its image. When powers of S form a basis, the coordinate statement is exactly the duplication of each off-diagonal unordered pair into its two ordered positions. □

The theorem is an upper bound on the common-column scalar image: extra public dependencies can only reduce that image further. Every concrete route checks the exact public rank before solving. The outer map cannot repair the loss because, after the relation, it sees only the already-equal polynomial values. The factorization itself uses only transposition symmetry. Odd characteristic is needed later for the familiar decomposition into symmetric and alternating tensors and for the polar-form analysis. The submitted nonsymmetrized $q=16$ construction is not covered.

3.3 Frobenius-distance normal form

The symmetric square has a canonical coordinate system that will be useful for the smaller adaptive cores. For $A=\mathbb{F}_{q^d}$ and $j>0$, define

$$\phi_j([x\otimes y]_{\mathrm{sym}}) = xy^{q^j}+x^{q^j}y, \qquad \phi_0([x\otimes y]_{\mathrm{sym}}) = xy.$$

If d is even and $j=d/2$, the value lies in $\mathbb{F}_{q^{d/2}}$.

Theorem 3.2 (Frobenius-distance normal form)**.** *There is an $\mathbb{F}_q$-linear isomorphism*

$$\mathrm{Sym}^2_{\mathbb{F}_q}(\mathbb{F}_{q^d}) \cong \begin{cases} A^{(d+1)/2}, & d \text{ odd},\\ A^{d/2}\oplus \mathbb{F}_{q^{d/2}}, & d \text{ even},\end{cases}$$

given by the maps ϕ_j for the distinct Frobenius distances. In particular,

$$\mathrm{Sym}^2_{\mathbb{F}_q}(\mathbb{F}_{q^4}) \cong \mathbb{F}_{q^4}\oplus\mathbb{F}_{q^4}\oplus\mathbb{F}_{q^2}, \tag{7}$$

$$\mathrm{Sym}^2_{\mathbb{F}_q}(\mathbb{F}_{q^2}) \cong \mathbb{F}_{q^2}\oplus\mathbb{F}_{q}. \tag{8}$$

Proof. Extend scalars to an algebraic closure and identify $A\otimes_{\mathbb{F}_q}\overline{\mathbb{F}}_q$ with a product of d copies indexed by $\mathbb{Z}/d\mathbb{Z}$. Frobenius cyclically shifts the primitive idempotents. The map ϕ_0 records the d diagonal unordered pairs. For $0<j<d/2$, the conjugates of ϕ_j record the orbit of unordered pairs $\{i,i+j\}$, an orbit of size d. If d is even and $j=d/2$, the pair is repeated after a half-turn, leaving an orbit of size $d/2$. These orbits partition all unordered pairs. The displayed map is therefore a coordinate permutation after scalar extension and hence an isomorphism over $\mathbb{F}_q$. □

For $\ell=4$, the three channel types have highest homogeneous forms of ordinary degrees 2, $q+1=20$, and $q^2+1=362$ over $A=\mathbb{F}_{19^4}$. For $\ell=2$, the complete quotient becomes two eigenblocks with bidegree families $(2,0)$, $(1,1)$, and $(0,2)$ after scalar extension.

4 The exact affine-column reduction

4.1 Imposing one common column variable

Write the stacked signature columns as

$$U = [u_0 \mid \cdots \mid u_{r-1}], \qquad u_j \in \mathbb{F}_q^{N_0}.$$

The attacker chooses multipliers $R_j \in A$ and offsets $v_j \in \mathbb{F}_q^{N_0}$, with $R_0 = 1$, and imposes

$$u_j = \Sigma_{R_j} u + v_j, \qquad 0 \le j < r. \tag{9}$$

This is a publicly checkable subset of possible signatures and makes no assumption about the honest signer.

Before substitution, the verifier feature vector has $N_{\mathrm{raw}} = m_1 \ell^2 r^2$ labeled coordinates. Let E be the public outer map after the fixed permutation that reconciles the official flattening order with the block-transposed order used here. For a fixed relation R, let R_R expand the remaining ordered power-pair features into the raw feature vector, and let $F_{R,v}$ collect the offset-linear coefficients. The complete substituted verifier has the form

$$E\big(R_R z_{\mathrm{ord}}(u) + F_{R,v} u\big) + c_{R,v} = t. \tag{10}$$

Put

$$E_R = E R_R, \qquad \Lambda_{R,v} = E F_{R,v}.$$

Products of two variable parts give the quadratics, products of one variable and one offset give the linear term, and products of two offsets give the constant.

Let $z_{\mathrm{sym}}(u)$ contain one coordinate for each unordered pair $0 \le a \le b < \ell$ in every base form, and let D duplicate its off-diagonal coordinates. By Theorem 3.1, $z_{\mathrm{ord}}(u) = D z_{\mathrm{sym}}(u)$. Define

$$Q_R = E_R D \in \mathbb{F}_q^{M \times K}, \qquad K = m_1 \binom{\ell+1}{2}. \tag{11}$$

The columns of Q_R span every output direction reachable by the quotient quadratics.

4.2 Canonical output coordinates

Choose W_R whose rows form a basis of the left kernel of Q_R, and put

$$C_{R,v} = W_R \Lambda_{R,v}. \tag{12}$$

Multiplication by W_R kills every quadratic feature and exposes ordinary affine constraints.

Lemma 4.1 (Canonical residual coordinates)**.** *Assume* $\operatorname{rank} Q_R = K$*. Choose a public left inverse* $G_R \in \mathbb{F}_q^{K \times M}$ *satisfying* $G_R Q_R = I_K$*. Then*

$$\begin{pmatrix} G_R \\ W_R \end{pmatrix} \in \mathbb{F}_q^{M \times M}$$

is invertible. Writing $b = t - c_{R,v}$*, the complete substituted verifier is equivalent to*

$$C_{R,v} u = W_R b, \qquad g_{R,v}(u) = G_R b, \qquad g_{R,v}(u) = z_{\mathrm{sym}}(u) + G_R \Lambda_{R,v} u. \tag{13}$$

The polar map of $g_{R,v}$ *is exactly that of* z_{sym}*. If* b *is uniform in* $\mathbb{F}_q^M$*, the two right-hand sides in* (13) *are independent and uniform.*

Proof. If a vector $y \in \mathbb{F}_q^M$ is killed by both G_R and W_R, then $W_R y = 0$ implies $y = Q_R a$ for some $a \in \mathbb{F}_q^K$. Applying G_R gives $a = G_R Q_R a = 0$, so $y = 0$. The stacked square matrix is therefore invertible. Applying its block rows to $Q_R z_{\mathrm{sym}}(u) + \Lambda_{R,v} u = b$ gives the two displayed systems, and invertibility gives the converse. Affine terms have zero polar. Finally, an invertible linear transformation sends a uniform vector to a uniform vector on the product of the two coordinate spaces. $\square$

Proposition 4.2 (Exact affine reduction)**.** *Let $\rho_Q = \operatorname{rank} Q_R$, $\lambda = \operatorname{rank} C_{R,v}$, and $\delta = M - \rho_Q - \lambda$. Then*

$$\operatorname{rank}[Q_R \ \Lambda_{R,v}] = \rho_Q + \lambda.$$

After row reduction, exactly δ independent affine consistency conditions remain on the target. For every target satisfying those conditions, Gaussian elimination leaves ρ_Q equations of degree at most two in $N_0 - \lambda$ variables, with exactly the same solution set as the complete substituted verifier.

If $\rho_Q = K$ and $\lambda = M - K$, every target passes the affine consistency test and the residual system consists of K equations on an affine translate of $\ker C_{R,v}$.

Proof. The row space of W_R is $\ker Q_R^{\mathsf{T}}$, so $\ker W_R = (\ker Q_R^{\mathsf{T}})^{\perp} = \operatorname{im} Q_R$. Hence W_R realizes the quotient map $\mathbb{F}_q^M \to \mathbb{F}_q^M / \operatorname{im} Q_R$. The image of the columns of $\Lambda_{R,v}$ in this quotient has dimension $\operatorname{rank}(W_R \Lambda_{R,v}) = \lambda$, proving the first identity.

The projected affine system is $C_{R,v} u = W_R b$. Its codomain has dimension $M - \rho_Q$ and its image has dimension λ, leaving $\delta = (M - \rho_Q) - \lambda$ target consistency equations. Whenever they hold, parameterize the affine solution space by $N_0 - \lambda$ free variables and substitute into a complementary set of ρ_Q rows. Every operation is invertible row reduction or exact affine elimination, so the solution set is unchanged. In the full-rank case, $C_{R,v}$ is surjective and $\delta = 0$. $\square$

Operational meaning. The attacker computes $Q_R, G_R, W_R, C_{R,v}$ and the canonical residual map from the supplied public key. The attacker may search public relations and offsets before an expensive solve, but may not replace an unfavorable supplied key. This release neither proves a success probability for that search nor charges it separately; any such search belongs in κ_{hom} or κ_{JG} for the selected route. If no tested relation passes, the conditional route has not been shown to apply to that key. Once the ranks pass, every solver root maps back through (9) to a candidate for the full original verifier. No equation has been dropped.

4.3 A Level-I dimension walkthrough

For $(v, o, \ell, r) = (28, 5, 4, 4)$, we have $n = 33$, $N_0 = 132$, $M = 80$, $m_1 = 5$, and $K = 50$. The common-column relation leaves 80 ordered quadratic labels, but the symmetry quotient leaves only 50 distinct quadratics. In the full-rank affine case, the other 30 output directions become linear equations and the full affine residual domain has dimension $132 - 30 = 102$. The complete-square theorem later chooses a fresh ordinary 50-dimensional slice inside that domain and solves all 50 residual equations together.

5 Stable slices, fresh coefficients, rejection, and targets

5.1 An unconditional stable kernel

Generic offsets can supply the missing affine output directions, but their linear terms may destroy the block grading used by multihomogeneous solvers. Choose one vector $w \in \mathbb{F}_q^{N_0}$ and multipliers

$\Gamma_0, \ldots, \Gamma_{r-1} \in A$, and set

$$v_j = \Sigma_{\Gamma_j} w. \tag{14}$$

Define the offset row space

$$U_w = \operatorname{span}_{\mathbb{F}_q}\{w^\mathsf{T}\Sigma_\xi P_i \Sigma_\eta : 1 \le i \le m_1,\ \xi, \eta \in A\}. \tag{15}$$

For a basis $1, S, \ldots, S^{d-1}$ of A, form the complete right-A orbit

$$O^F_{R,v} = \begin{pmatrix} F_{R,v} \\ F_{R,v}\Sigma_S \\ \vdots \\ F_{R,v}\Sigma_{S^{d-1}} \end{pmatrix}, \qquad \rho_F = \operatorname{rank} O^F_{R,v}. \tag{16}$$

Theorem 5.1 (Stable-lift kernel)**.** *Assume $S^\mathsf{T} = S$, $P_i^\mathsf{T} = P_i$, and $A \cong \mathbb{F}_{q^d}$. Under (14):*

(i) U_w is closed under right multiplication by A and $\dim_{\mathbb{F}_q} U_w \le m_1 d^2$;

(ii) every row of $F_{R,v}$ and $O^F_{R,v}$ lies in U_w;

(iii) the row space of $O^F_{R,v}$ is an A-subspace, so $d \mid \rho_F$;

(iv)

$$H^F_{R,v} = \ker O^F_{R,v}$$

is A-linear, satisfies $F_{R,v}H^F_{R,v} = 0$, and has

$$\dim_A H^F_{R,v} = n - \rho_F/d \ge n - m_1 d. \tag{17}$$

No generic orbit-rank equality is required. An unexpected dependency only enlarges the stable space.

Proof. The spanning family in (15) has $m_1 d^2$ members. For $\zeta \in A$,

$$(w^\mathsf{T}\Sigma_\xi P_i \Sigma_\eta)\Sigma_\zeta = w^\mathsf{T}\Sigma_\xi P_i \Sigma_{\eta\zeta},$$

so U_w is right-A stable. Every offset cross term has one occurrence of w; if it appears on the right, transpose the scalar and use symmetry and commutativity to put it on the left. Thus all rows of $F_{R,v}$ lie in U_w, and right stability places all orbit rows there as well.

Because (16) contains the complete right orbit of every row, its row space is an A-vector space, proving $d \mid \rho_F$. Its first block is $F_{R,v}$, so the kernel kills every offset-linear feature. If x lies in the kernel and $\zeta \in A$, each product $S^a\zeta$ is an $\mathbb{F}_q$-linear combination of the chosen basis powers; therefore the same orbit rows kill $\Sigma_\zeta x$. The kernel is A-linear. Rank–nullity over A gives (17). ☐

Since $C_{R,v} = W_R E F_{R,v}$, the stable kernel lies inside $\ker C_{R,v}$. Translation along a subspace $L \subseteq H^F_{R,v}$ changes the quotient quadratics but leaves every offset-linear feature constant.

5.2 Eigenblocks and multidegrees

Let Ω split the minimal polynomial of S, and let $e_0, \ldots, e_{\ell-1}$ be the primitive idempotents of $A \otimes_{\mathbb{F}_q} \Omega$. If L is A-linear of A-dimension s, then

$$L \otimes \Omega = L_0 \oplus \cdots \oplus L_{\ell-1}, \qquad \dim_\Omega L_a = s.$$

The unordered power-pair features are an invertible recombination of the unordered eigenblock features

$$x^\mathsf{T}\Sigma_{e_a} P_i \Sigma_{e_b} x, \qquad 0 \le a \le b < \ell.$$

They have multidegree $2\mathbf{e}_a$ when $a = b$ and $\mathbf{e}_a + \mathbf{e}_b$ when $a < b$. Translation creates lower-degree terms only in the same one or two blocks. One homogenizer per block therefore preserves these exact multidegrees. This is the grading used by the complementary streamed-XL branch.

Proof of the eigenblock statement. The power basis and the idempotent basis of $A \otimes \Omega \cong \Omega^\ell$ are related by an invertible Vandermonde matrix. Passing to the symmetric square gives an invertible change between unordered power-pair and unordered eigenblock coordinates. The idempotent e_a projects $L \otimes \Omega$ onto L_a. A diagonal feature uses one block twice; an off-diagonal feature is bilinear in two blocks. Translation and homogenization preserve the claimed supports. $\square$

5.3 Reserved-line decoupling and the structural preflight

Let $e_1, \ldots, e_n$ be the native A-coordinate basis in which the key-dependent symmetric public matrices are expanded. Fix a vinegar index j_0 and use the coordinate-aligned decomposition

$$\begin{aligned} &W = Ae_{j_0}, \qquad V' = \bigoplus_{\substack{j \text{ vinegar} \\ j \neq j_0}} Ae_j, \qquad O_{\text{pub}} = \bigoplus_{j \text{ oil}} Ae_j, \\ &V_{\text{pub}} = W \oplus V' \oplus O_{\text{pub}}, \quad \dim_A W = 1, \quad \dim_A V' = v - 1, \quad \dim_A O_{\text{pub}} = o. \end{aligned} \tag{18}$$

The actual SHAKE stream is deterministic once the public seed is fixed. For probability statements only, the *native-coordinate idealized XOF-transcript model* used below replaces the relevant stream bytes by independent uniform bytes and declares the native upper-triangular scalar entries of the key-dependent P_i to be their independent reductions modulo 19. This is not a theorem about a fixed SHAKE-generated key. The deterministic outer map E, the relation R, and the complete multiplier vector $\Gamma = (\Gamma_0, \ldots, \Gamma_{r-1})$ are fixed and conditioned upon. No A-linear change of basis is applied before invoking independence. A reduced byte has maximum atom

$$\rho_{\text{byte}} = \frac{14}{256},$$

because nine residues have fourteen preimages and ten have thirteen.

Lemma 5.2 (Coordinate-aligned reserved-line decoupling)**.** *Fix E, R, Γ, choose $0 \neq w \in W$, and use the structured offsets* (14)*. Then Q_R depends only on E and R, while $C_{R,v}$, $O^F_{R,v}$, $H^F_{R,v}$, and every deterministically tie-broken slice selected from them depend only on native entries having at least one index in W. In the native-coordinate idealized XOF-transcript model they are independent of the native $V' \times V'$ entries.*

Proof. Every structured offset is in W. Each offset-linear coefficient therefore contains one copy of w; if it occurs on the right, transpose the scalar and use symmetry to put it on the left. Right multiplication by A preserves the first coordinate block. Thus neither the orbit construction nor any deterministic linear-algebra operation on it reads a native $V' \times V'$ entry. The two sets of raw XOF symbols are disjoint. Coordinate alignment is essential: the biased byte-reduction distribution is not invariant under an arbitrary A-linear basis change. $\square$

Lemma 5.3 (Maximum-atom pivot bound)**.** *Let $X_1, \ldots, X_N$ be independent finite-field variables, each with maximum atom at most ρ_{byte}. If a matrix B has rank t, then for every c,*

$$\Pr[BX = c] \leq \rho_{\text{byte}}^t.$$

Proof. Choose t pivot columns. Condition on all nonpivot variables. Expose the pivot variables in echelon order; at each step at most one value can satisfy the next independent equation, and that value has probability at most ρ_{byte}. Multiplication gives the claim. $\square$

Put $\mathcal{Q}_R = \mathbb{F}_q^M / \operatorname{im} Q_R$. When $\operatorname{rank} Q_R = K$, $\dim_{\mathbb{F}_q} \mathcal{Q}_R = M - K$, and $C_{R,v}$ is a coordinate matrix for the induced linear map $u \mapsto W_R \Lambda_{R,v} u$ to $\mathcal{Q}_R$. Let $\mathbf{X}_{W,V'}$ be the vector of native independent public-matrix entries in the $W \times V'$ blocks. For $0 \neq \lambda \in \mathcal{Q}_R^\vee$, define B_λ by

$$\operatorname{vec}((\lambda \circ C_{R,v})|_{V'}) = B_\lambda \mathbf{X}_{W,V'} + b_\lambda.$$

Here B_λ and b_λ are fixed once E, R, Γ, λ and the native layout are fixed; they do not depend on the sampled entries in $\mathbf{X}_{W,V'}$. Scaling λ scales B_λ, so its rank depends only on $[\lambda]$. Put

$$t_{\text{src}} = \min_{0 \neq \lambda \in \mathcal{Q}_R^\vee} \operatorname{rank} B_\lambda.$$

For an A-line $\mathfrak{l} \subseteq W \oplus O_{\text{pub}}$, choose $0 \neq y_{\mathfrak{l}} \in \mathfrak{l}$ and write

$$O_{R,v}^F y_{\mathfrak{l}} = D_{\mathfrak{l}} \mathbf{X}_{\text{nat}} + e_{\mathfrak{l}}$$

in the complete vector $\mathbf{X}_{\text{nat}}$ of native independent public-matrix entries. The coefficient data $D_{\mathfrak{l}}, e_{\mathfrak{l}}$ are deterministic before those entries are sampled, and the rank of $D_{\mathfrak{l}}$ is independent of the chosen generator. Define

$$t_W = \operatorname{rank} D_W, \qquad t_{\text{oth}} = \min_{\substack{\mathfrak{l} \subseteq W \oplus O_{\text{pub}} \\ \dim_A \mathfrak{l} = 1,\ \mathfrak{l} \neq W}} \operatorname{rank} D_{\mathfrak{l}}.$$

These are exact symbolic coefficient-map ranks determined by the fixed relation, the complete offset pattern, and the native coordinate layout; they do not depend on the sampled values of the key coefficients. Define

$$\delta_{\text{src}} = \frac{q^{M-K} - 1}{q - 1} \rho_{\text{byte}}^{d(v-1)}, \tag{19}$$

$$\delta_{\text{proj}} = \rho_{\text{byte}}^{m_1 \binom{d+1}{2}} + \left(\frac{|A|^{1+o} - 1}{|A| - 1} - 1 \right) \rho_{\text{byte}}^{m_1 d^2}, \qquad |A| = q^d. \tag{20}$$

Theorem 5.4 (Conditional coefficient-preflight density)**.** *Fix the deterministic outer map E, a relation R, and the complete vector Γ. Assume the exact deterministic checks*

$$\operatorname{rank} Q_R = K, \qquad t_{\text{src}} \geq d(v-1), \qquad t_W \geq m_1 \binom{d+1}{2}, \qquad t_{\text{oth}} \geq m_1 d^2$$

pass. In the native-coordinate idealized XOF-transcript model,

$$\Pr[\operatorname{rank} C_{R,v} < M - K] \leq \delta_{\text{src}},$$

and

$$\Pr[H_{R,v}^F \cap (W \oplus O_{\text{pub}}) \neq 0] \leq \delta_{\text{proj}}.$$

Consequently full affine rank and injectivity of $H_{R,v}^F \to V'$ hold jointly with probability at least $1 - \delta_{\text{src}} - \delta_{\text{proj}}$. For the six official $\ell = 4$ dimension tuples, direct substitution gives $\delta_{\text{src}} + \delta_{\text{proj}} < 2^{-209}$. The coefficient-map ranks are fixed-layout symbolic checks; the realized quotient, affine, and intersection ranks are exact public checks on a fixed key.

Proof. If $\operatorname{rank} C_{R,v} < M-K$, some nonzero $\lambda \in \mathcal{Q}_R^\vee$ annihilates $C_{R,v}$. In particular, $(\lambda \circ C_{R,v})|_{V'} = 0$. For fixed $[\lambda]$, this is an affine system of rank at least $d(v-1)$ in the independent native variables $\mathbf{X}_{W,V'}$; Theorem 5.3 bounds its probability by $\rho_{\mathrm{byte}}^{d(v-1)}$. Scalar multiples define the same event, and $\mathbb{P}(\mathcal{Q}_R^\vee)(\mathbb{F}_q)$ has $(q^{M-K}-1)/(q-1)$ points. A union bound proves (19).

By Theorem 5.1, $H_{R,v}^F$ is A-linear. Thus a nonzero intersection with $W \oplus O_{\mathrm{pub}}$ contains an A-line $\mathfrak{l}$. For $\mathfrak{l} = W$, the coefficient-rank hypothesis and Theorem 5.3 give probability at most $\rho_{\mathrm{byte}}^{m_1\binom{d+1}{2}}$. For every other A-line the analogous bound is $\rho_{\mathrm{byte}}^{m_1 d^2}$. The $(1+o)$-dimensional A-space $W \oplus O_{\mathrm{pub}}$ contains $(|A|^{1+o}-1)/(|A|-1)$ lines, exactly one of which is W; a union bound gives (20). Finally, the kernel of the projection $H_{R,v}^F \to V'$ is precisely $H_{R,v}^F \cap (W \oplus O_{\mathrm{pub}})$. A final union bound combines the two bad events without assuming independence. Direct substitution gives the displayed uniform numerical bound. ☐

No tail of Γ is implicit in this statement: the concrete route records all r entries and runs the displayed checks for the vector above. This release does not assert that it passes them for every fixed key. The six $\ell = 4$ ledgers below are conditional on a successful structural preflight.

5.4 The guaranteed fresh ordinary square

Let

$$D_0 = V' \cap \ker C_{R,v}.$$

The intersection inequality gives

$$\dim_{\mathbb{F}_{19}} D_0 \geq 4(v-1) - (M-K). \tag{21}$$

For all six $\ell = 4$ rows this lower bound is at least K:

Parameters	$\dim D_0$ lower bound	K
$(28,5,4,4)$, $(28,4,4,5)$	78	50
$(40,7,4,4)$	114	70
$(38,5,4,5)$	118	70
$(50,9,4,4)$	142	90
$(52,6,4,6)$	150	90

Consequently the attacker can choose, from cross data alone, an ordinary K-dimensional slice inside D_0. Its internal $V' \times V'$ coefficients remain fresh for the spectrum theorem. This simple dimension fact is the reason the complete-square route applies to every $\ell = 4$ row, including $(38,5,4,5)$; it does not require the smaller structured stable kernel to have A-dimension $K/4$.

5.5 Passing the block-symmetry rejection rule

Version 2.3 rejects a square $\ell = 4$ signature if any block is symmetric and rejects an $\ell = 2$ signature if more than $\lfloor n/4 \rfloor$ blocks are symmetric [4]. Rectangular $\ell = 4$ blocks are not subject to the square-block rule.

For a square $\ell = 4$ relation $R = (R_0, R_1, R_2, R_3)$, define the local skew map

$$\mathcal{S}_R : \mathbb{F}_q^4 \longrightarrow \bigwedge^2 \mathbb{F}_q^4, \qquad \mathcal{S}_R(x)_{a,c} = (R_c x)_a - (R_a x)_c.$$

For any offsets, one signature block is symmetric exactly when $\mathcal{S}_R(x)$ equals a fixed offset-dependent vector. The public relation $R_j = S^j$ for the official matrix has $\operatorname{rank} \mathcal{S}_R = 4$. Thus each block has at most one rejected common-column value, and the accepted density in the full common-column space is at least

$$\alpha_{4,n} = (1 - 19^{-4})^n. \tag{22}$$

The accepted-root theorem later uses only this global density; it assumes no independence from the MQ system.

For $\ell = 2$, the zero-offset relation used in the main theorem gives a block

$$\begin{pmatrix} x & 0 \\ y & 0 \end{pmatrix},$$

which is symmetric exactly when $y = 0$. The exact accepted density is derived in (39). A deterministic alternative is also available: for the official matrix

$$S = \begin{pmatrix} 1 & 2 \\ 2 & 15 \end{pmatrix}, \qquad R_1 = 9I + 10S = \begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix},$$

choose offsets with $(v_1)_{b,0} - (v_0)_{b,1} = 1$ in every block. Then each block has fixed nonzero skew 1, so every root passes the rejection rule. This fixed-skew route is a useful separate algebraic check but is not needed for the selected-degree-free headline.

5.6 Exactly uniform official targets

The probability theorems use a uniform target over $\mathbb{F}_{19}^M$. The official binary-to-field conversion has a small modulo bias. The attacker removes it without changing the verifier. Here the target hash/XOF chunks are modeled as independent uniform bit strings, separately from the native-coordinate key-expansion model above. The pinned conversion uses each full eight-byte chunk for fifteen base-19 digits and a final partial chunk for the remaining digits. If a uniform b-bit chunk X is to supply a base-19 digits, retain the message–salt input only when

$$X < \left\lfloor \frac{2^b}{19^a} \right\rfloor 19^a. \tag{23}$$

Conditioned on this event, $X \bmod 19^a$ is uniform. Applying the test to each chunk makes the retained official target exactly uniform, and an invertible output change preserves uniformity. Exact evaluation over the nine public Version 2.3 lengths gives a minimum acceptance probability $0.1524622985\ldots$, and hence at least 0.152462. When target consistency requires more than the 128-bit salt space, the existential chosen-message attacker varies a message prefix as well as the salt. Target generation occurs before nonlinear solving and adds to, rather than multiplies, the solve cost.

5.7 The exact first and second moments

The accepted-root theorem rests on an exact collision count. We state it here so that the probability argument is self-contained.

Theorem 5.5 (Root-collision identity)**.** *Let U be an $\mathbb{F}_q$-space, let $C : U \to \mathbb{F}_q^c$ be surjective, let $L \subseteq \ker C$ have dimension h, and let $g : U \to \mathbb{F}_q^K$ have degree at most two. Sample independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, then choose a uniform affine coset $\mathcal{A}$ of L inside $C^{-1}(d)$. Put*

$$Z = |\{x \in \mathcal{A} : g(x) = z\}|, \qquad \mu = q^{h-K}.$$

For $0 \neq y \in L$, define

$$T_y(x) = g(x+y) - g(x) - g(y) + g(0)$$

and let $\kappa(y)$ be one if $-(g(y) - g(0)) \in \operatorname{im} T_y$ and zero otherwise. Then

$$\mathbb{E}Z = \mu, \tag{24}$$

$$\mathbb{E}[Z(Z-1)] = \mu \sum_{0 \neq y \in L} \kappa(y) q^{-\operatorname{rank} T_y} \leq \mu \sum_{0 \neq y \in L} q^{-\operatorname{rank} T_y}. \tag{25}$$

The domain of T_y is the full space U because the experiment averages over the affine right-hand side as well as the coset.

Proof. Sampling d and then a uniform L-coset in $C^{-1}(d)$ is exactly uniform over all cosets of L in U. Every point of U therefore lies in the sampled coset with probability $q^{h-\dim U}$, and its image matches the independent target with probability q^{-K}. Summing over all points gives (24).

For an ordered pair of distinct roots in one coset, write the second point as $x + y$ with $0 \neq y \in L$. The collision condition is

$$g(x+y) = g(x) \quad \Longleftrightarrow \quad T_y(x) = -(g(y) - g(0)).$$

It has $\kappa(y) q^{\dim U - \operatorname{rank} T_y}$ solutions in x. Divide the total number of ordered collisions by the number $q^{\dim U - h} q^K$ of target–coset pairs to obtain (25). □

6 Full-domain spectrum and conditional square-system recovery

The conditional recovery route has a deliberately simple primary theorem. For a key of any published parameter shape that passes the stated exact preflights, the symmetry quotient and affine split leave a square residual system: K equations on a K-dimensional public slice. The full-domain spectrum theorem supplies accepted nonsingular roots, and the conditional finite-cardinality homotopy adaptation recovers them only under $\mathsf{H}_{\mathrm{hom}}$. No selected degree, chosen Jacobian core, family-count sweep, or exact-corank prediction appears in this direct route.

Layer	Mathematical statement	Status
Exact algebra	Public-matrix symmetry factors the ordered feature map through $\operatorname{Sym}^2(A)$; the output split retains every verifier equation and reconstructs every candidate.	Exact for the scalar common-column map of a fixed key and relation passing the stated public ranks.
Root density	A full-domain directional-rank spectrum bounds both collisions and the singular locus of the restricted Jacobian.	Theorem in the idealized XOF-transcript model, or an exact projective-rank certificate for a fixed key.
Complete-square recovery	Restrict the full K-coordinate residual map to a public K-dimensional slice and solve all K equations together. Every candidate is checked by the original verifier.	Conditional on $\mathsf{H}_{\mathrm{hom}}$ as well as the stated public and spectrum events; no production-size solver execution is supplied.
Certified optimization	An exact determinant preflight may replace the complete square by a smaller channel core; failure invokes the complete square or a lower-work orbit fallback.	A certified branch optimization, not an assumption about the public key.

The direct conditional theorem is the conceptual headline. The adaptive cores improve its arithmetic margins, but they are not needed to state the recovery route for the nine parameter shapes under the common hypotheses, including $\mathsf{H}_{\mathrm{hom}}$. This separation makes the logical claim independent of the optimization machinery; it does not convert the nominal ledger into a demonstrated break.

The numerical statements below are unit-leading AXN operation ledgers. They are not wall-clock timings, low-memory theorems, or complete attack circuits. They make no polynomial-time claim: the displayed multihomogeneous Bézout quantities are exponential, and κ_{hom} remains visible.

6.1 The spectrum required by the accepted-root theorem

Let U be a finite-dimensional $\mathbb{F}_q$-space, let $C : U \to \mathbb{F}_q^c$ be surjective, let $g : U \to \mathbb{F}_q^K$ have degree at most two, and let $L \subseteq \ker C$ have dimension h. Write $g = q_2 + q_1 + q_0$ by homogeneous degree and define, for $0 \neq y \in L$,

$$T_y : U \longrightarrow \mathbb{F}_q^K, \qquad T_y(x) = g(x+y) - g(x) - g(y) + g(0), \tag{26}$$

$$\bar{\eta}_L^U = \sum_{0 \neq y \in L} q^{-\operatorname{rank} T_y}. \tag{27}$$

The superscript matters. The accepted-root experiment first samples a uniform affine right-hand side for C and then a uniform L-coset in that fiber. Its collision calculation averages over the whole domain U. Consequently the relevant derivative in (26) has domain U, not $\ker C$. Restricting T_y to $\ker C$ gives a different, generally larger quantity that is useful for fixed-fiber uniqueness but is not needed here.

Let $X \subseteq U$ be the assignments accepted by the original verifier and put $\alpha = |X|/|U|$. A trial samples independent uniform $d \in \mathbb{F}_q^c$ and $z \in \mathbb{F}_q^K$, and then a uniform affine coset of L inside $C^{-1}(d)$.

Theorem 6.1 (Accepted nonsingular root, full-domain form)**.** *Assume $\bar{\eta}_L^U \leq \eta$ and $\alpha > \eta/(q-1)$. A trial contains an accepted root x satisfying*

$$\operatorname{rank} D_L g(x) = h, \qquad D_L g(x)[y] = T_y(x) + q_1(y),$$

with probability at least

$$p_{\mathrm{ns}} = q^{h-K} \frac{a^2}{a+\eta}, \qquad a = \alpha - \frac{\eta}{q-1}. \tag{28}$$

No independence among acceptance, singularity, the target, and the quadratic map is assumed.

Proof. Let Y count accepted roots in the sampled target–coset pair at which $D_L g$ has rank h, and let Z count all roots. For a projective direction $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$, the condition $D_L g(x)[y] = 0$ is an affine system in x whose linear part is T_y. It therefore holds on at most a $q^{-\operatorname{rank} T_y}$ fraction of U. Union-bound over projective directions to obtain

$$\frac{|\{x : \operatorname{rank} D_L g(x) < h\}|}{|U|} \leq \frac{1}{q-1} \sum_{0 \neq y \in L} q^{-\operatorname{rank} T_y} \leq \frac{\eta}{q-1}.$$

Every point of U is selected by the random fiber–coset with probability $q^{h - \dim U}$ and matches the independent target with probability q^{-K}. Put $\mu = q^{h-K}$ and $a = \alpha - \eta/(q-1)$. Hence $\mathbb{E}Y \geq \mu a$. The ordered-pair collision identity gives $\mathbb{E}[Z(Z-1)] \leq \mu\eta$, and $Y(Y-1) \leq Z(Z-1)$. Therefore

$$\mathbb{E}[Y^2] = \mathbb{E}Y + \mathbb{E}[Y(Y-1)] \leq \mathbb{E}Y + \mu\eta.$$

The function $x \mapsto x^2/(x+\mu\eta)$ is increasing for $x \geq 0$. Cauchy–Schwarz therefore gives

$$\Pr[Y > 0] \geq \frac{(\mathbb{E}Y)^2}{\mathbb{E}Y + \mu\eta} \geq \mu \frac{a^2}{a+\eta},$$

which proves (28). The denominator is strictly smaller than $1+\eta$ whenever $a < 1$. □

The following lemma improves the seeded theorem because it prices exactly the full-domain spectrum used above. Let $\rho_{\text{byte}} = 14/256$ be the maximum atom of one public-expansion byte reduced modulo 19.

Lemma 6.2 (Full-domain fresh-spectrum theorem)**.** *Assume q is odd and $A = \mathbb{F}_{q^d}$. Work in the native-coordinate idealized XOF-transcript model specified in Section 5: the scalar entries in the fresh symmetric-matrix region are independent reductions of independent uniform bytes, each having maximum atom $\rho_{\text{byte}} = 14/256$, while all disjoint coefficient data are conditioned upon. Let L be fixed from coefficient data disjoint from that fresh region. Suppose every $0 \neq y \in L$ has nonzero projection $\bar{y}$ to a fresh A-space D, and every nonzero residual-output combination contains a nonzero coefficient $c \in \mathrm{Sym}^2_{\mathbb{F}_q}(A)$ in at least one fresh public base form. If*

$$\dim_{\mathbb{F}_q} D = d + t,$$

then for all $0 \neq y \in L$ and $0 \neq b \in \mathbb{F}_q^K$,

$$\Pr[b \in \ker T_y^{\mathsf{T}}] \leq \rho_{\text{byte}}^t. \tag{29}$$

Consequently

$$\mathbb{E}[\bar{\eta}_L^U] \leq (q^h - 1)q^{-K} + (q^h - 1)(1 - q^{-K})\rho_{\text{byte}}^t, \tag{30}$$

$$\Pr\Big[\bar{\eta}_L^U > (q^h - 1)q^{-K} + \varepsilon\Big] \leq \frac{(q^h - 1)(1 - q^{-K})\rho_{\text{byte}}^t}{\varepsilon}. \tag{31}$$

For the reserved-line construction one may take $D = V' \cong A^{v-1}$ and $t = d(v-2)$. For the zero-offset construction on the full public-vinegar summand one may take $D \cong A^v$ and $t = d(v-1)$.

Proof. Fix $y \neq 0$ and $b \neq 0$, and choose a fresh public base form on which the corresponding symmetric-square coefficient c is nonzero. If $c = \sum_\nu a_\nu \odot b_\nu$, the coefficient vector of the fresh symmetric matrix in the scalar derivative $b \cdot T_y(x)$ is

$$\Xi_{c,\bar{y}}(x) = \sum_\nu \big((a_\nu \bar{y}) \odot (b_\nu x) + (b_\nu \bar{y}) \odot (a_\nu x)\big).$$

Its kernel is contained in the A-line $A\bar{y}$. Indeed, if $x \notin A\bar{y}$, project to the cross summand between the disjoint lines $A\bar{y}$ and Ax. After identifying both lines with A, the projection is the ordinary symmetrization of c, which is nonzero because symmetrization is injective on $\mathrm{Sym}^2(A)$ in odd characteristic. Thus the coefficient map has rank at least $\dim_{\mathbb{F}_q} D - d = t$.

The event $b \in \ker T_y^{\mathsf{T}}$ forces a rank-t affine system in independent fresh symbols. Sequentially condition on nonpivot symbols and apply the maximum-atom bound to the pivots, proving (29). Finally,

$$q^{-\operatorname{rank} T_y} = q^{-K} |\ker T_y^{\mathsf{T}}| = q^{-K}\left(1 + \sum_{b \neq 0} \mathbf{1}_{\{b \in \ker T_y^{\mathsf{T}}\}}\right).$$

Taking expectations gives $q^{-K} + (1 - q^{-K})\rho_{\text{byte}}^t$. Sum over $y \neq 0$ to obtain (30). The excess above the unavoidable baseline $(q^h - 1)q^{-K}$ is nonnegative, so Markov's inequality gives (31). The two displayed choices of D have dimensions $d(v-1)$ and dv, respectively. □

We henceforth use the nearly ideal threshold

$$\boxed{\eta_{128}(h,K) = (q^h - 1)q^{-K} + 2^{-128}.} \tag{32}$$

This is not a heuristic claim of perfect rank. It is a theorem-backed allowance of only 2^{-128} above the exact full-rank baseline.

Corollary 6.3 (Numerical spectrum tails)**.** *For the cost-driving slices,* (31) *with* $\varepsilon = 2^{-128}$ *gives the following base-two failure exponents. The entry in each cell is the weaker official shape when two shapes share a level.*

Route	*Level I*	*Level III*	*Level V*
$\ell = 4$ *fast channel core*	-138.12	-237.86	-371.14
$\ell = 4$ *direct complete square*	-95.64	-178.39	-294.67
$\ell = 2$ *one-eigenblock core*	-130.17	-263.46	-396.74
$\ell = 2$ *complete two-block core*	-62.21	-161.50	-260.80

Thus, in that idealized transcript model, even the weakest headline event fails with probability below 2^{-62}*. A fixed key can instead publish its exact projective rank histogram, which certifies* (32) *without the idealized XOF-transcript model.*

6.2 Core-complete recovery and the adaptive theorem

After scalar extension, partition the residual equations into multidegree families $\mathcal{F}_1, \dots, \mathcal{F}_r$. A count vector $\mathbf{k} = (k_1, \dots, k_r)$ chooses k_j linear combinations from family j, with $\sum_j k_j = h$. It is *core-complete* if every point at which the full restricted Jacobian has rank h has a nonzero h-row minor with a count vector in the chosen family.

Lemma 6.4 (Random projection within equation families)**.** *Let* $J_j \in E^{m_j \times h}$ *contain the Jacobian rows in family* j*. Suppose some choice of* k_j *rows from each family stacks to a nonzero* $h \times h$ *minor. If* $P_j \in E^{k_j \times m_j}$ *is uniform, then*

$$\Pr\left[\det\begin{pmatrix} P_1 J_1 \\ \vdots \\ P_r J_r \end{pmatrix} = 0\right] \le \frac{h}{|E|}.$$

Proof. The determinant is a polynomial of total degree at most h in the entries of the P_j. It is not identically zero because coordinate-selector matrices recover the assumed minor. Apply Schwartz–Zippel. □

Lemma 6.5 (Conservative well-separating-form success)**.** *For a square multihomogeneous core with Bézout number* B*, represented over a finite field* E *large enough for the finite-cardinality start system, the well-separating-form construction has a set of at most* B^2 *nonzero forbidden coefficient vectors. A uniform linear form therefore satisfies every separation condition with probability at least*

$$\sigma_E(B) = 1 - \frac{B^2}{|E|}.$$

Conditional on this event and $\mathsf{H}_{\mathrm{hom}}$*, the finite-cardinality homotopy algorithm returns a zero-dimensional parametrization containing every nonsingular core root, with one-sided error.*

Proof. The finite-cardinality homotopy construction reduces failure of its well-separating element to vanishing on a collection of at most B^2 nonzero vectors over E. For each fixed nonzero vector, a uniform coefficient vector annihilates it with probability at most $|E|^{-1}$. The union bound proves the displayed probability. We deliberately retain this theorem-level B^2 bound: ordinary pairwise separation alone would involve fewer hyperplanes, but it does not by itself verify every condition required by the cited homotopy parametrization. □

Theorem 6.6 (Conditional core-complete Las Vegas recovery)**.** *Assume $\mathsf{H}_{\mathrm{hom}}$ for every invoked core, (32), and let $\mathcal{P}$ be a finite core-complete family. Run every count vector in $\mathcal{P}$ with fresh within-family projections and a valid finite-cardinality homotopy field. Filter every parametrization through all residual equations, base-field descent, the signature rejection rule, affine reconstruction, and the original verifier. If $W_{\mathbf{k}}$ is the unit-leading invocation cost before root, projection, and separator success factors, then*

$$\mathbb{E}W \le \frac{\sum_{\mathbf{k}\in\mathcal{P}} W_{\mathbf{k}}}{p_{\mathrm{ns}}\ \min_{\mathbf{k}}(1-h/|E_{\mathbf{k}}|)\ \min_{\mathbf{k}}\left(1-\frac{B_{\mathbf{k}}^2}{|E_{\mathbf{k}}|}\right)}. \tag{33}$$

The algorithm is Las Vegas and assumes no selected Macaulay degree, exact corank, unique root, or predetermined nonzero Jacobian core. Its actual work is κ_{hom} times the displayed ledger.

Proof. By Theorem 6.1, a trial contains an accepted full-Jacobian root with probability at least p_{ns}. At such a root, core completeness supplies a count vector with a nonzero minor. The family-projection lemma preserves it with the displayed conditional probability, and the separating form succeeds with the second displayed conditional probability. The corresponding homotopy invocation then contains the root in its output. Running every count vector guarantees that this invocation is present. Summing invocation costs and inverting the common success lower bound proves (33). Complete public filtering prevents an invalid output. □

A *fast-core preflight* is an exact public test proving that the highest homogeneous part of the selected core's Jacobian determinant is not the zero polynomial. In the applications below, evaluating that highest part at one fixed public point suffices.

Theorem 6.7 (Adaptive certified-core fallback)**.** *Fix a public construction for which a common structural and spectrum event $\mathcal{S}$ makes a core-complete fallback valid. Suppose:*

1. *on passing an exact public fast-core preflight, the fast route has expected unit-leading work at most W_f;*
2. *on failing it, the core-complete fallback has expected work at most W_F;*
3. *under the idealized XOF-transcript distribution, the preflight fails with probability at most δ, while $\Pr[\mathcal{S}^c] \le \varepsilon$.*

For every fixed key in $\mathcal{S}$, the algorithm "test, then choose fast or fallback" is selected-degree-free and fixed-core-free. Under the idealized transcript distribution, its conditional expected work obeys

$$\mathbb{E}[W \mid \mathcal{S}] \le W_f + \min\left\{1, \frac{\delta}{1-\varepsilon}\right\}(W_F - W_f)_+. \tag{34}$$

If an independent exact structural preflight succeeds with probability p_{str}, the transcript-model-normalized ledger divides the right side by $p_{\mathrm{str}}(1-\varepsilon)$. Without independence one may instead use any proved lower bound on $\Pr[\mathcal{S}]$.

Proof. On a fixed key, passing the preflight is a certificate, not a probabilistic premise. If it passes, the fast theorem applies; if it fails, core completeness applies. Thus no preferred core is assumed. For the expected work, let F be preflight failure. Then

$$\Pr[F \mid \mathcal{S}] \le \frac{\Pr[F]}{\Pr[\mathcal{S}]} \le \min\{1, \delta/(1-\varepsilon)\}.$$

Conditioning on the two branches gives $(1-q)W_f + qW_F = W_f + q(W_F - W_f)$ when $W_F \ge W_f$; if $W_F < W_f$, always using the fallback only helps, yielding the positive-part form. The last statement is the ordinary normalization by the density of idealized transcripts that reach the solver. □

7 Constructive arithmetic and the homotopy ledger

The primary ledger below represents pointwise extension-field products in the tower basis throughout and charges the displayed recurrence for each product. It does not charge conversion of official or manuscript-basis coefficients, nor the bulk transforms used by the homotopy evaluation/interpolation schedule. Those conversions and transforms, together with additions, inversions, factorization, control, memory, and all remaining hidden constants, stay in κ_{hom}.

Theorem 7.1 (Constructive scalar netlist and tower recurrences)**.** *The circuit model assigns unit cost to two-input AND, XOR, and XNOR gates, with free constants, wires, and fan-out. For canonical five-bit representatives, the following circuit-size assignments are constructive:*

	addition	*negation*	*subtraction*	*multiplication*
$\mathbb{F}_{19}$	42	16	58	150
$\mathbb{F}_{19^2}$	84	32	116	692
$\mathbb{F}_{19^4}$	168	64	232	2628.

The emitted 150*-gate scalar netlist is correct on all* $19^2 = 361$ *canonical input pairs. The* 692 *and* 2628 *extension-field multiplication figures are constructive composition counts from the displayed primitive recurrences; the release does not emit complete netlists for those two multipliers. The tower representations are*

$$\mathbb{F}_{19^2} = \mathbb{F}_{19}[u]/(u^2+1), \qquad \mathbb{F}_{19^4} = \mathbb{F}_{19^2}[v]/(v^2-(1+u)).$$

They are exactly isomorphic to the manuscript basis $\mathbb{F}_{19}[t]/(t^4-t-1)$ *via*

$$u = 1 + 2t + 15t^2 + 5t^3, \qquad v = 4 + 13t + 13t^2 + t^3,$$

and the basis $(1, u, v, uv)$ *has determinant* $16 \ne 0$ *modulo* 19*.*

Proof. For a canonical scalar $x = (x_0, \dots, x_4)$, the circuit first converts to balanced signed magnitude. On the valid inputs $0, \dots, 18$, set

$$a = x_1x_3, \quad b = x_2x_3, \quad s = a \oplus b \oplus ax_2 \oplus x_4,$$

$$(m_0, m_1, m_2, m_3) = (x_0 \oplus s, x_1 \oplus s, x_2, x_3 \oplus b).$$

A direct truth-table check gives $s = 0, m = x$ for $x \le 9$ and $s = 1, m = 19 - x$ for $10 \le x \le 18$. The two decoders use 18 gates. An exact four-by-four magnitude product together with the output sign reaches gate 80. Since the magnitude product is at most 81, write it as $\ell + 32h$ with $h \in \{0, 1, 2\}$, replace $32h$ by $13h \bmod 19$, fold once, and reduce the reachable five-bit values; this reaches gate 130. A 20-gate signed recoding maps r to either r or $-r \bmod 19$. The released netlist contains all 150 gates and a separately implemented evaluator checks every canonical pair and canonical output.

For $u^2 = -1$, Karatsuba uses three scalar products, two input sums, and shared output negations, giving

$$5 \cdot 42 + 2 \cdot 16 + 3 \cdot 150 = 692.$$

At the second tower level, three $\mathbb{F}_{19^2}$ products cost $3 \cdot 692$. The two input sums cost $2 \cdot 84$, multiplication by $1 + u$ costs one scalar addition and subtraction, the low output costs one $\mathbb{F}_{19^2}$ addition, and the high output forms $p_0 + p_1$ once and subtracts it from p_2. Hence

$$2 \cdot 84 + 3 \cdot 692 + (42 + 58) + 84 + (84 + 116) = 2628.$$

Direct polynomial arithmetic verifies $u^2 = -1$, $v^2 = 1 + u$, the displayed coordinates, and the basis determinant. $\square$

For a core over $A \in \{\mathbb{F}_{19^2}, \mathbb{F}_{19^4}\}$ with Bézout number B and start-node requirement D_{start}, let g_A be 692 or 2628 and define

$$\Gamma_A(B, D_{\text{start}}) = \min_{\substack{e \geq 1:\ |A|^e \geq D_{\text{start}},\ |A|^e > B^2 \\ 2e-1 \leq |A|}} \frac{(2e-1)g_A}{1 - B^2/|A|^e}. \tag{35}$$

Evaluation/interpolation at $2e - 1$ base-field points gives the number of charged pointwise multiplications in the numerator; the fixed transforms themselves remain in κ_{hom}. Theorem 6.5 gives the denominator. The numerical artifact exhausts all admissible e.

8 The direct square-system theorem

The following specialization is the main conceptual simplification. It uses all residual equations on a square slice, so no Jacobian-core choice or family-count enumeration remains.

Theorem 8.1 (Direct conditional square-system recovery)**.** *Retain the setting of Theorem 6.1, assume H_{hom}, and suppose $h = K$. Restrict the complete residual map g to the public affine K-dimensional slice and solve all K equations together. On the event $\bar{\eta}_L^U \leq \eta$, a trial contains an accepted root at which the Jacobian of this very system is nonsingular with probability at least*

$$p_\square = \frac{a^2}{a + \eta}, \qquad a = \alpha - \frac{\eta}{q - 1}.$$

If every equation has ordinary degree at most two, then

$$B_\square = 2^K, \qquad B_\square^+ = 2^K \left(1 + \frac{K}{2}\right).$$

Let

$$D_K = \binom{K+1}{2}, \qquad L_\square = D_K + 2K(D_K + K + 1), \qquad H_\square = L_\square + 2K + K^2.$$

Under H_{hom} and over any field satisfying the displayed finite-cardinality conditions, the unit-leading work is

$$W_\square = \frac{\Gamma_A(2^K, 2K)}{p_\square} 2^{2K} \left(1 + \frac{K}{2}\right) H_\square K. \tag{36}$$

The algorithm is Las Vegas after complete public filtering and assumes no selected Macaulay degree, exact corank, unique root, preferred core, or orbit sweep.

Proof. With $h = K$, Theorem 6.1 applies directly to the Jacobian of the complete residual system; no projection to a subcore is made. The ordinary total-degree Bézout number of K quadratics in K variables is 2^K. In the companion homotopy coefficient, selecting the auxiliary term from no factor contributes 2^K, while selecting it from one of the K degree-two factors contributes 2^{K-1}; hence $B^+_\square = 2^K + K2^{K-1}$. The displayed straight-line program first forms all D_K quadratic monomials and then evaluates the K affine quadratics, charging a coefficient multiplication and addition for every possible term. The conditional homotopy adaptation and Theorem 6.5 give (36). Every point returned by the parametrization is checked against the full residual system, affine reconstruction, rejection rule, and original verifier. $\square$

Corollary 8.2 (Conditional direct-square routes for the six $\ell = 4$ shapes)**.** *For every official $\ell = 4$ shape, any public key passing the exact quotient and affine-rank preflights leaves a fresh ordinary K-dimensional slice in the public-vinegar region. Under the applicable spectrum inequality and $\mathsf{H}_{\mathrm{hom}}$, applying Theorem 8.1, embedding the $\mathbb{F}_{19}$ coefficients into $\mathbb{F}_{19^2}$, and charging the constructive* 692*-gate multiplication recurrence gives normalized exponents*

$$\boxed{142.60696, \qquad 185.45088, \qquad 227.56041}$$

on the worse shape at Levels I, III, and V. The corresponding nominal headroom below $143, 207, 272$ *is*

$$\boxed{0.39304, \qquad 21.54912, \qquad 44.43959}$$

bits before κ_{hom}. In particular, the two direct Level-I comparisons survive only for multipliers below approximately 1.313157 *and* 1.313691*, respectively; a factor of two erases both crossings. Thus these are nominal multiplication-schedule comparisons, not implementation margins.*

Proof. The manuscript's fresh-slice inequality gives $\dim(V' \cap \ker C_{R,v}) \geq K$ for all six rows, so a cross-data-selected ordinary K-slice exists. The full-domain spectrum theorem applies with $t = 4(v-2)$ and the threshold $\eta_{128}(K, K)$. Substitute $K = 50, 70, 90$ in (36). Exact enumeration of the separator extension degree selects $e = 13, 17, 22$ over $\mathbb{F}_{19^2}$. The machine-readable ledger conditions on the exact structural preflight, normalizes only the certified spectrum event, and charges the conservative square-row acceptance lower bound. $\square$

9 The zero-offset $\ell = 2$ theorem

For the three official $\ell = r = 2$ rows put

$$A = \mathbb{F}_{19^2}, \qquad m = m_1 = o, \qquad M = 4m, \qquad K = 3m.$$

Use the zero-offset column relation

$$u_0 = u, \qquad u_1 = 0. \tag{37}$$

Proposition 9.1 (Exact zero-offset target split)**.** *Let Q_R be the quotient-output matrix for (37). The route applies to a fixed key only after the exact public preflight* $\operatorname{rank} Q_R = K$*. No probability claim replaces that fixed-key test for the three $\ell = 2$ rows. After a passing preflight, choose a left inverse G_R and a full-row-rank matrix W_R spanning the left kernel of Q_R. The complete substituted verifier is equivalent to*

$$W_R t = 0, \qquad z_{\mathrm{sym}}(u) = G_R t. \tag{38}$$

For a uniform official target, consistency has probability $19^{-(M-K)} = 19^{-m}$, and conditioned on it the residual target is uniform in $\mathbb{F}_{19}^K$. Chosen-message and salt inputs can therefore be filtered before solving. If one deliberately charges 2^{32} AXN operations per retained exactly uniform target, the additive target-generation exponents are

$$16\log_2 19 + 32 = 99.96684, \quad 24\log_2 19 + 32 = 133.95026, \quad 32\log_2 19 + 32 = 167.93368.$$

Proof. Zero offsets make every affine and constant contribution vanish, so the substituted verifier is $Q_R z_{\mathrm{sym}}(u) = t$. Applying the invertible output change with block rows G_R, W_R gives (38). An invertible linear transformation preserves uniformity and separates the quotient and consistency coordinates. Target generation occurs before the nonlinear solve, so its work adds to rather than multiplies the solve ledger. □

In one block, (37) has matrix $\left(\begin{smallmatrix} x & 0 \\ y & 0 \end{smallmatrix}\right)$ and is symmetric exactly when $y = 0$. The $n = v + o$ block conditions use disjoint coordinates. Thus the exact density accepted by the Version 2.3 rule is

$$\alpha_n = 19^{-n} \sum_{j=0}^{\lfloor n/4 \rfloor} \binom{n}{j} 18^{n-j}. \tag{39}$$

Choose all solver slices inside the full public-vinegar summand A^v, independently of its fresh internal symmetric coefficients. By Theorem 6.2, the fresh exponent is $t = 2(v-1)$.

9.1 The complete two-block core

Take an A-linear slice of dimension $s = 3m/2$. Its base-field dimension is $h = 2s = K$, and the complete quotient system has m equations in each bidegree family

$$(2,0), \qquad (1,1), \qquad (0,2).$$

It is therefore square and core-complete without selecting or discarding any residual equation.

Theorem 9.2 (Conditional complete two-block $\ell = 2$ recovery)**.** *On the event $\bar{\eta}_L^U \le \eta_{128}(K,K)$, a consistent target–coset trial contains an accepted full-Jacobian root with probability at least*

$$p_{\mathrm{comp}}^{(2)} = \frac{a_2^2}{a_2 + \eta_{128}(K,K)}, \qquad a_2 = \alpha_n - \frac{\eta_{128}(K,K)}{18}. \tag{40}$$

The exact multihomogeneous quantities are

$$B_m^{(2)} = 2^{2m}\binom{m}{m/2}, \tag{41}$$

$$(B_m^{(2)})^+ = B_m^{(2)}\left(1 + 2m + \frac{m^2}{m+2}\right). \tag{42}$$

Assuming $\mathsf{H}_{\mathrm{hom}}$, finite-cardinality homotopy over $\mathbb{F}_{19^2}$, Frobenius descent, and complete public filtering form a selected-degree-free, fixed-core-free Las Vegas recovery route under the displayed event. The normalized primary exponents, including the additive target filter and the 2^{-128} spectrum normalization, are unit-leading multiplication-schedule exponents before the omitted $\log_2 \kappa_{\mathrm{hom}}$ term:

$$132.97067, \qquad 183.79463, \qquad 233.84399$$

for Levels I, III, and V.

Proof. Apply Theorem 6.1 with $h = K$ and the exact accepted density (39). Over the splitting field, each base form supplies one equation in each of the two diagonal families and the cross family. The Bézout number is

$$[t_0^s t_1^s](2t_0)^m (t_0 + t_1)^m (2t_1)^m.$$

The cross factor must contribute $m/2$ powers to each block, proving (41). In the companion homotopy coefficient, the term with no auxiliary variable contributes B. Deleting a diagonal factor contributes $2^{2m-1}(\binom{m}{m/2} + \binom{m}{m/2+1})$, while deleting a cross factor contributes B. Summing and using $\binom{m}{m/2+1}/\binom{m}{m/2} = m/(m+2)$ proves (42). The exact separator factor is (35); every later operation is an exact filter. $\square$

Put

$$\begin{aligned} D_s &= \binom{s+1}{2}, & C_s &= (s+1)^2, \\ L_{\mathrm{comp}} &= 2D_s + s^2 + 4m(D_s + s + 1) + 2mC_s, \\ H_{\mathrm{comp}} &= L_{\mathrm{comp}} + 6m + K^2. \end{aligned} \tag{43}$$

The unit-leading work on a good spectrum event is

$$W_{\mathrm{comp}}^{(2)} = \frac{\Gamma_{\mathbb{F}_{19^2}}(B_m^{(2)}, 2s)}{p_{\mathrm{comp}}^{(2)}} B_m^{(2)} (B_m^{(2)})^+ H_{\mathrm{comp}} K. \tag{44}$$

Divide by the exact spectrum-event density and add the target-generation ledger.

9.2 The faster Level-I core and its exact fallback

For $s \leq m$, select s diagonal equations and solve one eigenblock of A^s. A descended full root reconstructs the other block by Frobenius. The core is a square system of s ordinary quadratics, with

$$B_s = 2^s, \qquad B_s^+ = 2^s(1 + s/2).$$

At a fixed nonzero public point, the fresh leading row of each selected public form is surjective onto $(A^s)^\vee$. Sequential exposure therefore gives

$$p_{\mathrm{Jac}}^{(2)}(s) \geq \prod_{j=1}^{s} (1 - \rho_{\mathrm{byte}}^{2j}) > 0.9970003329. \tag{45}$$

A nonzero leading determinant has degree at most s over A, so its singular point density is at most $s/19^2$. With $\eta_s = \eta_{128}(2s, 3m)$, a target–coset trial has an accepted full root nonsingular for this core with probability at least

$$p_s = 19^{2s-3m} \max\left\{a_s - \eta_s, \frac{a_s^2}{a_s + \eta_s}\right\}. \tag{46}$$

where $a_s = \alpha_n - s/19^2$. These are respectively the singleton and second-moment lower bounds after subtracting the core-singular locus.

At Level I take $s = m = 16$. Run the exact leading-determinant preflight. If it passes, solve the one-eigenblock core; if it fails, solve the complete two-block system of Theorem 9.2. By Theorem 6.7, this is fixed-core-free by construction and has normalized exponent

$$\boxed{132.18807}. \tag{47}$$

The diagonal-only and complete-only exponents in the reported ledger are 132.18928 and 132.97067. The adaptive value is slightly smaller than the first rounded figure because the common spectrum normalization is charged once and the exact branch mixture is performed before taking logarithms. At Levels III and V, the complete core is already faster.

10 An explicit Boolean-ledger Level-I route via Just Guess

The homotopy ledger deliberately leaves κ_{hom} explicit. This section asks whether the row that determines the adaptive Level-I maximum, $(v, o, \ell, r) = (48, 16, 2, 2)$, admits a more concrete estimate. We combine the exact zero-offset target split of Theorem 9.1 with the underdetermined-MQ transformation of May, Ostuzzi, and Ressler [20]. Their analysis reduces a random underdetermined MQ system to guessed variables, a short tree of univariate quadratics, a linear solve, and check equations. It explicitly uses a random-system regularity model for the linearization step. We retain that premise rather than turning it into an unconditional statement.

Write the quotient map in its two Frobenius channels as

$$D : A^{64} \longrightarrow A^{16}, \qquad H : A^{64} \longrightarrow \mathbb{F}_{19}^{16}, \qquad A = \mathbb{F}_{19^2}.$$

Here D is the diagonal channel and H is the Hermitian cross channel. The retained consistent target is (t_D, t_H). We additionally reject the event $t_D = 0$, of probability 19^{-32} under the exact uniform target sampler. Conditioned on consistency and $t_D \neq 0$, the pair (t_D, t_H) remains exactly uniform over the retained target set.

The two channel families are distinct linear images of the same native byte-reduced public coefficients. They are therefore *not independent*. The argument below uses an exact conditional atom bound for the official $\ell = 2$ channel transform instead.

10.1 The streamed diagonal solver

Apply Just Guess to the $m = 16$ equations $D(x) = t_D$ in $n = 64$ variables over A, with

$$k = 5, \qquad p = 6, \qquad m - k - p = 5.$$

The published structural inequalities hold:

$$2k + p = m, \qquad 2(n - m) = 96 \geq 6(2m - 2k - p - 1) = 90,$$
$$n - 1 = 63 \geq (m - k - p - 1)(m - k + 2) = 52.$$

The public transformation gives coordinates $(u, z) \in A^{16} \times A^{48}$. Fixing z changes only linear and constant terms in the transformed square system. We stream an A^8 family of z assignments. For each assignment, the solver enumerates all $|A|^5$ guesses, follows every returned quadratic branch, solves the remaining 5×5 linear system, and checks the five withheld equations. Surviving points are tested against $H(x) = t_H$, the format rule, affine reconstruction, and the unmodified verifier.

Let $\mathcal{R}$ denote the event that this computation supplies at least $19^{16}/4$ distinct sign-canonical diagonal roots. This is a concrete, transcript-checkable candidate-abundance premise motivated by, but not proved by, the published Just Guess random-system analysis. No reduced-size or production transcript in the release establishes $\mathcal{R}$. We write κ_{JG} for the expected multiplier needed to obtain $\mathcal{R}$. It includes transformation and linearization retries, candidate-abundance failures, the negligible format and $t_D = 0$ losses, and any candidate-processing cost beyond the explicit reserve below. Unlike κ_{hom}, it does not hide a generic soft-O algebraic solver, but it remains an unmeasured premise. If $\mathcal{R}$ is unreachable for a fixed system and transformation family, then $\kappa_{\text{JG}} = \infty$ and the comparison gives no crossing.

10.2 Conditional cross-channel filtering

For $x \in A^{64}$ put $\Psi(x) = xx^{19\top}$. If $\Psi(x) = \Psi(y)$ for nonzero x, y, then $y = \lambda x$ with $\lambda\lambda^{19} = 1$. If also $D(x) = D(y) = t_D \neq 0$, then $\lambda^2 = 1$, so $y = \pm x$. Retaining one representative of each sign pair therefore makes the cross tensors distinct.

Lemma 10.1 (Exact conditional channel atom)**.** *Use the native-coordinate idealized XOF-transcript model, in which one field symbol has raw integer weight* 14 *for residues* $0, \ldots, 8$ *and* 13 *for residues* $9, \ldots, 18$*. Condition on the complete diagonal-channel coefficient family* D*. For every native coefficient block and every nonzero* $\mathbb{F}_{19}$*-linear functional of its remaining Hermitian coefficient, every conditional atom has probability at most*

$$\rho_{\text{cond}} = \frac{49}{829} < 0.059108. \tag{48}$$

Conditioned on D*, different native blocks and different base forms remain independent.*

Proof. Diagonalize the official matrix $S = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 15 \end{smallmatrix}\right)$ over $A = \mathbb{F}_{19}[u]/(u^2+1)$. Its eigenvalues are $8 \pm 2u$, with eigenvectors $(1, 13 \pm u)$. For a native symmetric diagonal block $\left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$, the diagonal coefficient and Hermitian coefficient are

$$D = (a + 7b + 16c) + (2b + 7c)u, \qquad H = a + 7b + 18c.$$

For an off-diagonal native block $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, they are

$$D = (a + 13b + 13c + 16d) + (b + c + 7d)u,$$

$$H = (a + 13b + 13c + 18d) + (18b + c)u.$$

These maps act separately on disjoint native blocks. Exact enumeration of all 19^3 diagonal tuples with their product weights gives maximum conditional atom $49/829$. Exact enumeration of all 19^4 off-diagonal tuples and all 20 projective nonzero linear functionals on the A-valued Hermitian coefficient gives the smaller maximum $169246/2971565$. The artifact reproduces both rational values directly from the displayed formulae. Conditioning separately on the D-image of each independent native block preserves blockwise and base-form independence. $\square$

Lemma 10.2 (Conditional cross-channel second moment)**.** *Let* C *be any set of* $T = c\,19^{16}$ *sign-canonical solutions of* $D(x) = t_D$*, where* $t_D \neq 0$ *and* $0 < c \leq 1$*. Condition on the entire diagonal channel and on any solver transcript determined by it. For a uniform independent target* t_H*,*

$$\Pr[\exists x \in C : H(x) = t_H \mid D] \geq \frac{c}{1 + c(19\rho_{\text{cond}})^{16}}. \tag{49}$$

For $c = 1/4$*, this lower bound is* $0.0961344407\ldots$*.*

Proof. Let $N = |\{x \in C : H(x) = t_H\}|$. Uniformity of t_H gives $\mathbb{E}N = c$. For distinct $x, y \in C$, the Hermitian tensors differ. For each of the sixteen independent base forms, the equality $H_i(x) = H_i(y)$ is a nonzero linear equation in the Hermitian block coefficients. Condition on all blocks except one on which this functional is nonzero. By Theorem 10.1, its conditional probability is at most ρ_{cond}. Independence across the sixteen base forms gives

$$\Pr[H(x) = H(y) \mid D] \leq \rho_{\text{cond}}^{16}.$$

Consequently

$$\mathbb{E}N(N-1) \leq T(T-1)19^{-16}\rho_{\text{cond}}^{16} < c^2(19\rho_{\text{cond}})^{16}.$$

Cauchy–Schwarz gives $\Pr[N > 0] \geq (\mathbb{E}N)^2/\mathbb{E}[N^2]$, proving the claim. $\square$

10.3 Finite Boolean-gate instantiation

We use the constructive 692-gate multiplication recurrence and 84-gate adder count over A. A monic quadratic over A is solved by a fixed Tonelli–Shanks schedule. The artifact checks all 361 possible discriminants and obtains

$$C_{\text{root}} = 16{,}886 \quad \mathsf{AXN} \text{ gates.} \tag{50}$$

Instantiating the published expected Just Guess operation formulas at $(n, m, k, p) = (64, 16, 5, 6)$ gives, per guess,

2684.089 extension multiplications, 2621.087 extension additions, 6 quadratic-root calls.

Rounding upward and adding one million gates for comparisons, address logic, and branch control gives

$$C_{\text{guess}} = 3{,}179{,}584 \quad \mathsf{AXN} \text{ gates.} \tag{51}$$

We declare a reserve of 2^{40} gates per z assignment for candidate filtering and verifier calls. This is a ledger allowance, not a theorem that it dominates every possible candidate population; any excess is included in κ_{JG}. Target sampling, the public transformation, and all rank preflights are polynomial-sized additive terms.

Proposition 10.3 (Conditional finite-gate Level-I sensitivity)**.** *Under the published Just Guess expected-operation model, the Boolean ledger for one A^8-streaming trial is*

$$W_{\text{JG}}^{\text{trial}} = |A|^{8+5} C_{\text{guess}} + |A|^8 2^{40} < 2^{132.046523}. \tag{52}$$

On event $\mathcal{R}$, Theorem 10.2 with $c = 1/4$ gives an expected cross-channel restart multiplier below 10.403*. Consequently the success-adjusted Boolean-gate ledger is*

$$\log_2 W_{\text{JG}} < 135.425325 + \log_2 \kappa_{\text{JG}}. \tag{53}$$

It is below the Level-I reference 143 *whenever*

$$\log_2 \kappa_{\text{JG}} < 7.574675. \tag{54}$$

*For orientation only, a factor-*16 *regularity reserve gives an exponent below* 139.425325 *and more than* 3.574675 *bits of remaining margin. A separate capped-tree schedule follows at most* 63 *quadratic nodes and* 64 *leaves per guess and aborts any branch whose residual linear system is not unique. Its trial exponent is below* 137.713504*; after the same cross-channel restart it is below* $141.092307 + \log_2 \kappa_{\text{JG}}^{\text{cap}}$*. Thus this strictly bounded-work sensitivity remains below* 143 *when* $\log_2 \kappa_{\text{JG}}^{\text{cap}} < 1.907693$*. It is reported as a sensitivity check, not as a proof of the capped candidate event.*

Proof. There are $|A|^8$ streamed assignments and $|A|^5$ guesses per assignment. Substituting the scalar circuit, tower recurrence counts, and quadratic-root schedule into the published expected-operation formulas gives (52); the artifact recomputes the integer ledger before taking logarithms. On $\mathcal{R}$, take any $19^{16}/4$ sign-canonical candidates. The reciprocal of (49) is $10.4020993\ldots$, adding 3.378803 bits. The remaining regularity, transformation, candidate-collection, and excess filtering multipliers are, by definition, included in κ_{JG}. Every candidate is checked by the original verifier, so all failure modes are one-sided. $\square$

Interpretation. This proposition replaces the homotopy soft-O multiplier on the row setting the adaptive Level-I maximum by (i) an explicit Boolean ledger for the published expected-operation schedule and (ii) the specific, unestablished but transcript-checkable event that the Just Guess run yields enough diagonal roots. It does not provide a complete circuit for every control path or prove candidate abundance. A complete, instrumented fixed-key implementation would estimate κ_{JG}. The two $\ell = 4$ Level-I rows still retain κ_{hom} in their homotopy ledgers.

11 Adaptive $\ell = 4$ recovery

Let $A = \mathbb{F}_{19^4}$ and let $L \cong A^s$ be the public slice. Under the Frobenius-distance normal form, each public base form supplies an A-valued degree-two channel and an A-valued degree-20 channel. Write their highest homogeneous parts as

$$F_{i,0}^{\mathrm{top}}(x) = C_{i,0}(x,x), \qquad F_{i,1}^{\mathrm{top}}(x) = C_{i,1}(x, x^{19}).$$

Lemma 11.1 (Fresh channel decomposition and row surjectivity)**.** *Restriction of a fresh symmetric $\mathbb{F}_{19}$-bilinear form to the injected public-vinegar copy of L, followed by the Frobenius-distance transform, is surjective onto*

$$\mathrm{Sym}^2_A(L^\vee) \oplus \mathrm{Bil}_A(L, L^{(19)}; A).$$

For every fixed $0 \neq x_\star \in L$, the map from one public form's fresh coefficients to the pair of leading Jacobian rows

$$(DF_{i,0}^{\mathrm{top}}(x_\star), DF_{i,1}^{\mathrm{top}}(x_\star)) \in L^\vee \oplus L^\vee$$

is surjective over A.

Proof. After extending scalars, the four diagonal blocks of a symmetric form form one Frobenius orbit and descend to an arbitrary symmetric A-bilinear form; the four adjacent blocks form a second orbit and descend to an arbitrary bilinear form between L and its Frobenius twist. The orbit coefficient summands are disjoint, so their direct-sum coefficient map is surjective. Formal differentiation gives

$$DF_{i,0}^{\mathrm{top}}(x_\star)[h] = 2C_{i,0}(x_\star, h), \qquad DF_{i,1}^{\mathrm{top}}(x_\star)[h] = C_{i,1}(h, x_\star^{19}).$$

Evaluation of arbitrary bilinear forms is surjective. For the symmetric case, given $\lambda \in L^\vee$, choose $\phi(x_\star) = 1$ and use

$$\frac{1}{2}(\phi \otimes \lambda + \lambda \otimes \phi - \lambda(x_\star)\phi \otimes \phi).$$

□

Choose a whole coordinates of the degree-two channel and b whole coordinates of the degree-20 channel, with $a + b = s$. The determinant preflight is exact on the public key. For its idealized-transcript probability we use the following conservative bound, which does not assume that an invertible linear change preserves independence under the slightly biased modulo-19 distribution.

Lemma 11.2 (Projective Jacobian-preflight density)**.** *At the fixed public preflight point, let $J \in A^{s \times s}$ be the selected leading-Jacobian matrix, where $A = \mathbb{F}_{19^4}$. Then*

$$\Pr[\det J = 0] \le \frac{|A|^s - 1}{|A| - 1} \rho_{\mathrm{byte}}^{4s}. \tag{55}$$

For the cost-minimizing values $s = 10, 14, 18$, the failure bounds are at most $3.55350 \cdot 10^{-5}$, $6.56029 \cdot 10^{-5}$, and $1.21113 \cdot 10^{-4}$, respectively.

Proof. If J is singular, some projective direction $[y] \in \mathbb{P}^{s-1}(A)$ lies in its right kernel. Fix such a direction. The joint channel-surjectivity statement above implies that forcing every selected leading row to annihilate y imposes s independent equations over A, hence $4s$ independent affine equations over $\mathbb{F}_{19}$, on fresh public-expansion symbols. The maximum-atom bound gives probability at most ρ_{byte}^{4s}. There are $(|A|^s - 1)/(|A| - 1)$ projective directions. Union-bound over them. This bound is weaker than a sequential full-rank product would be under a uniform field distribution, but it is valid for the actual biased product distribution used in the idealized XOF-transcript model. ☐

Conditional on passing the preflight, put

$$a_{a,b} = \alpha - \frac{a + 19b}{19^4}.$$

The sharpened core-root probability is

$$p_{a,b,s} = 19^{4s-K} \frac{a_{a,b}^2}{a_{a,b} + \eta_{128}(4s, K)}. \tag{56}$$

Its exact one-block Bézout data are

$$B = 2^a 20^b, \qquad B^+ = B(1 + a/2 + b/20).$$

With

$$\begin{aligned} L_{a,b,s} = \mathbf{1}_{a>0} &\left[\binom{s+1}{2} + 2a \left(\binom{s+1}{2} + s + 1 \right) \right] \\ &+ \mathbf{1}_{b>0} \left[6s + s^2 + 2b(s^2 + 2s + 1) \right], \end{aligned}$$

under H_{hom} the unit-leading good-key work, before the multiplier κ_{hom}, is

$$\frac{\Gamma_{\mathbb{F}_{19^4}}(B, 2a + 20b)}{p_{a,b,s}} BB^+ (L_{a,b,s} + 2a + 20b + s^2) s. \tag{57}$$

Exhaustive enumeration of all 35, 63, and 99 admissible profiles gives

$$(s; a, b) = (10; 5, 5), \quad (14; 7, 7), \quad (18; 9, 9)$$

at Levels I, III, V. Conditional on the exact structural preflight and the stated spectrum/model events, and assuming H_{hom}, the worse-shape fast-core ledger exponents before $\log_2 \kappa_{\text{hom}}$ are 126.260, 167.038, and 207.273.

For the simplest fixed-core-free algorithm, a failed preflight invokes the direct ordinary square of Theorem 8.2. Applying Theorem 6.7 gives:

Level	parameters	fast profile	fast conditional	adaptive mixture	headroom
I	$(28, 5, 4, 4)$	$(10; 5, 5)$	126.260	128.246	14.754
I	$(28, 4, 4, 5)$	$(10; 5, 5)$	126.260	128.246	14.754
III	$(40, 7, 4, 4)$	$(14; 7, 7)$	167.038	171.617	35.383
III	$(38, 5, 4, 5)$	$(14; 7, 7)$	167.038	171.616	35.384
V	$(50, 9, 4, 4)$	$(18; 9, 9)$	207.273	214.558	57.442
V	$(52, 6, 4, 6)$	$(18; 9, 9)$	207.273	214.557	57.443

Every exponent in this table assumes H_{hom} and excludes the additive term $\log_2 \kappa_{\text{hom}}$. The table needs no family-count sweep. Only the direct and adaptive routes whose complete numerical inputs are regenerated by the artifact are reported. No sixteen-orbit or output-size-optimized numerical branch is asserted.

12 All-nine consequences

Theorem 12.1 (One conditional recovery theorem for every published shape)**.** *For every published $q = 19$ Version 2.3 parameter shape, a public key passing the stated exact rank preflights, satisfying the applicable spectrum inequality (whether established by the idealized model event or by a fixed-key certificate), and satisfying $\mathsf{H}_{\mathrm{hom}}$ has a selected-degree-free Las Vegas recovery route obtained by solving a square residual system. Its expected target-generation interpretation additionally uses the stated idealized transcript model. The theorem specifies a conditional forgery wrapper under these conditions; it does not assert that they hold for every official key or that the released artifact has executed the production-size solver. It uses no selected Macaulay degree, exact-corank prediction, unique-root assumption, predetermined Jacobian core, family-count pattern, or orbit enumeration. The unit-leading homotopy ledgers are:*

Level	*parameters*	*complete square*	*exponent*	*headroom*
I	$(28, 5, 4, 4)$	*ordinary* 50×50 *square*	142.607	0.393
I	$(48, 16, 2, 2)$	*complete two-block square*	132.971	10.029
I	$(28, 4, 4, 5)$	*ordinary* 50×50 *square*	142.606	0.394
III	$(40, 7, 4, 4)$	*ordinary* 70×70 *square*	185.451	21.549
III	$(72, 24, 2, 2)$	*complete two-block square*	183.795	23.205
III	$(38, 5, 4, 5)$	*ordinary* 70×70 *square*	185.450	21.550
V	$(50, 9, 4, 4)$	*ordinary* 90×90 *square*	227.560	44.440
V	$(96, 32, 2, 2)$	*complete two-block square*	233.844	38.156
V	$(52, 6, 4, 6)$	*ordinary* 90×90 *square*	227.559	44.441

The worst rows at Levels I, III, and V have exponents

$$\boxed{142.60696, \qquad 185.45088, \qquad 233.84399}. \tag{58}$$

Thus even this deliberately uniform multiplication-schedule ledger is nominally below the $143, 207, 272$ *references at all three levels before κ_{hom}. This numerical comparison is not a complete-gate break.*

Proof. The $\ell = 2$ rows are Theorem 9.2; the $\ell = 4$ rows are Theorem 8.2. Both routes use the same accepted-root theorem, exact separator ledger, and complete public filtering. The displayed worst values are the levelwise maxima. □

Theorem 12.2 (Optimized adaptive conditional ledger)**.** *Under the same structural, spectrum, transcript, and $\mathsf{H}_{\mathrm{hom}}$ conditions, an exact public determinant preflight can replace the direct square by a smaller core without introducing a preferred-core premise: failure invokes the direct complete square. The resulting all-nine table is*

Level	*parameters*	*adaptive selected-degree-free route*	*exponent*	*headroom*
I	$(28, 5, 4, 4)$	*channel core, then ordinary square*	128.246	14.754
I	$(48, 16, 2, 2)$	*one eigenblock, then two-block square*	132.188	10.812
I	$(28, 4, 4, 5)$	*channel core, then ordinary square*	128.246	14.754
III	$(40, 7, 4, 4)$	*channel core, then ordinary square*	171.617	35.383
III	$(72, 24, 2, 2)$	*complete two-block square*	183.795	23.205
III	$(38, 5, 4, 5)$	*channel core, then ordinary square*	171.616	35.384
V	$(50, 9, 4, 4)$	*channel core, then ordinary square*	214.558	57.442
V	$(96, 32, 2, 2)$	*complete two-block square*	233.844	38.156
V	$(52, 6, 4, 6)$	*channel core, then ordinary square*	214.557	57.443

Taking the worst row at each level gives

$$\boxed{132.18807, \qquad 183.79463, \qquad 233.84399}. \tag{59}$$

The corresponding nominal headroom is approximately

$$\boxed{10.81193, \qquad 23.20537, \qquad 38.15601} \text{ bits.}$$

For example, sufficient conditions rounded downward from the exact ledger are

$$\log_2 \kappa_{\text{hom}}^{(I)} < 10.811927, \qquad \log_2 \kappa_{\text{hom}}^{(III)} < 23.205374, \qquad \log_2 \kappa_{\text{hom}}^{(V)} < 38.156013$$

at Levels I, III, and V. No particular value of these multipliers is asserted.

Proof. Use the exact branch rule of Theorem 6.7. For $\ell = 2$, the Level-I fast branch is the certified one-eigenblock core and the fallback is the complete two-block square; the complete square is already faster at Levels III and V. For $\ell = 4$, use Theorem 11.2 and fall back to Theorem 8.2. The machine-readable ledger mixes branch work before taking logarithms, charges the union of the relevant spectrum tails, and is conditional on the exact structural preflight; only the common spectrum-event density is normalized probabilistically. □

Scope of the arithmetic unit. The primary tables charge the emitted, exhaustively validated $\mathbb{F}_{19}$ multiplier and the constructive tower-recurrence counts for pointwise products over $\mathbb{F}_{19^2}$ and $\mathbb{F}_{19^4}$ represented in the tower basis. Conversion of official or manuscript-basis coefficients, the bulk extension-field evaluation/interpolation maps and their transforms, together with additions, inversions, factorization, control, filtering, memory traffic, and implementation constants, remain inside κ_{hom}. A narrower variable-product-only normalization is retained only in the artifact and is not used for any theorem headline.

13 Complementary streamed-XL and low-state frontiers

The complete-square homotopy route supplies a conditional time bound under H_{hom}, but it does not supply a low-memory implementation. This section records a complementary branch whose matrices can be streamed with linear vector state. Its correctness is exact once a finite quotient certificate is produced; its numerical exponent retains a production selected-degree premise. The two branches therefore answer different questions rather than competing for one headline.

13.1 Four-block special XL

For an A-linear $\ell = 4$ slice of A-dimension s, scalar extension gives four blocks of s variables. Each unordered pair of blocks carries m_1 independent quadratic equations. After adding one homogenizer per block, the Hilbert series used by the published special-XL model is

$$H_{m_1,s}(\mathbf{t}) = \frac{\prod_{a=1}^{4}(1 - t_a^2)^{m_1} \prod_{1 \le a < b \le 4}(1 - t_a t_b)^{m_1}}{\prod_{a=1}^{4}(1 - t_a)^{s+1}}. \tag{60}$$

For a multidegree $\mathbf{d} = (d_1, \ldots, d_4)$, the monomial-column count is

$$M_s(\mathbf{d}) = \prod_{a=1}^{4} \binom{s + d_a}{d_a}. \tag{61}$$

A negative coefficient of (60) is the selected-degree trigger in the cited model [8]. It predicts that a planted instance has the finite quotient needed for extraction; the theorem below makes the operator and state accounting exact but does not prove that production instances reach such a quotient at that degree.

The exact row count is

$$\begin{aligned} R_s(\mathbf{d}) = m_1 \sum_{a=1}^{4} \binom{s+d_a-2}{d_a-2} \prod_{c\neq a} \binom{s+d_c}{d_c} \\ + m_1 \sum_{1\le a<b\le 4} \binom{s+d_a-1}{d_a-1}\binom{s+d_b-1}{d_b-1} \prod_{c\notin\{a,b\}} \binom{s+d_c}{d_c}, \end{aligned} \tag{62}$$

where a binomial coefficient with a negative lower argument is zero.

Proposition 13.1 (Streaming the special-XL operator)**.** *Let $A_{\mathbf{d}}$ be the Macaulay matrix at multidegree $\mathbf{d}$, with $M = M_s(\mathbf{d})$ columns and $R = R_s(\mathbf{d})$ rows. Every row and transpose-row product can be generated from the public quadratic coefficients using $O(s^2)$ scratch space.*

Over an extension $E/\mathbb{F}_{19^4}$, choose a random diagonal $\Delta \in E^{R\times R}$ and put

$$B = A_{\mathbf{d}}^{\mathsf{T}} \Delta A_{\mathbf{d}}.$$

Then

$$\Pr[\ker B \neq \ker A_{\mathbf{d}}] \le M/|E|.$$

A product by B is evaluated by one pass over the streamed rows and uses $O(M)$ resident field-vector state. Scalar Wiedemann over $E = \mathbb{F}_{19^8}$ requires fewer than

$$16R(s+1)^2M \tag{63}$$

signed $\mathbb{F}_{19^4}$ multiply–accumulates per trial. A conservative cubic preconditioner over $\mathbb{F}_{19^{12}}$ gives the Level-I bound $32R(s+1)^2M$.

Proof. The two sums in (62) count multipliers of the diagonal and off-diagonal equation families. A homogenized quadratic row has at most $(s+1)^2$ terms, so rows and transpose products are generated on demand.

Factor $A_{\mathbf{d}} = UV$ at rank r. If $U^{\mathsf{T}}\Delta U$ is nonsingular, then $\ker(A_{\mathbf{d}}^{\mathsf{T}}\Delta A_{\mathbf{d}}) = \ker V = \ker A_{\mathbf{d}}$. By Cauchy–Binet, $\det(U^{\mathsf{T}}\Delta U)$ is a nonzero multilinear polynomial of total degree r in the diagonal entries. Schwartz–Zippel gives failure at most $r/|E| \le M/|E|$.

Streaming first computes the row inner product and then accumulates the scaled row. In the quadratic extension, the sparse dot product and accumulation use at most four base-extension products per row term and the diagonal multiplication uses three more. Fewer than $3M$ normal-operator products generate the scalar Wiedemann sequence and reconstruct one null vector. Rounding the resulting constant gives (63); the cubic-extension calculation is analogous. □

13.2 Certified extraction beyond exact corank one

A one-dimensional Macaulay kernel is convenient but is not required for soundness.

Theorem 13.2 (Recovery from a finite commuting border quotient)**.** *Fix a target and slice over a finite field Ω, and let $I \subseteq \Omega[x_1, \ldots, x_h]$ be the ideal of the residual equations. Let $\mathcal{B}$ be a finite* *divisor-closed monomial set containing* 1*. Suppose the Macaulay row space supplies, for every border monomial* $\beta \in \partial\mathcal{B}$*, a certified relation*

$$g_\beta = \beta - \sum_{b\in\mathcal{B}} c_{\beta,b} b \in I.$$

Construct the formal multiplication matrices $M_1, \ldots, M_h$ *on* $\Omega\mathcal{B}$*. If they commute, then the border relations generate a zero-dimensional ideal* $J \subseteq I$*,* $\mathcal{B}$ *is a basis of* $\Omega[x]/J$*, and every root of* I *occurs among the roots enumerated from the multiplication tables. FGLM conversion costs* $O(h|\mathcal{B}|^3)$ *field operations once the tables are available. Complete residual and public verification removes every extra point.*

Proof. Commutativity makes $x_i \mapsto M_i$ an algebra homomorphism. Divisor closure identifies each basis monomial with its coordinate vector, and every border relation acts as zero. The border-division theorem therefore gives unique normal forms in $\operatorname{span}_\Omega \mathcal{B}$, so the relations form a border basis of a zero-dimensional ideal J. Because every relation is a Macaulay row combination from I, we have $J \subseteq I$ and hence $V(I) \subseteq V(J)$. Standard FGLM conversion and finite-field factorization enumerate the geometric points of J [15, 16]. □

A quotient of dimension larger than one is therefore a correctness fallback, not a free extension of the scalar-Wiedemann exponent. Obtaining all border relations, block-nullspace work, and resident multiplication-table state must fit the displayed ledger or be charged separately.

13.3 Three-plus-one completion

The fourth eigenblock can be completed linearly after solving the first three. Let E_{ab} be the m_1 equations supported on blocks a, b.

Theorem 13.3 (Exact three-plus-one completion)**.** *Suppose a certified first-stage solver returns a finite set containing every root of the six families* E_{ab} *with* $1 \le a \le b \le 3$*. For each partial root* $p = (x_1, x_2, x_3)$*, substitute into* E_{14}, E_{24}, E_{34} *and obtain*

$$A_p x_4 = b_p, \qquad A_p \in \Omega^{3m_1\times s}.$$

Discard inconsistent systems. If $\operatorname{rank} A_p = s$*, solve for the unique fourth block and retain it exactly when* $E_{44} = 0$*. If a consistent system has rank below* s*, declare the trial incomplete unless a separately priced procedure enumerates all completions. Whenever no incomplete case occurs, the retained set contains every root of the complete four-block system. Final descent and verification make the procedure Las Vegas.*

Proof. Any complete root restricts to a first-stage root. At that partial root its true fourth block satisfies the displayed linear system. On a trial that declares no incomplete case, every consistent system encountered has full column rank. Hence the true fourth block is the unique completion, and the remaining diagonal family checks it. Conversely every retained point satisfies all ten families. □

13.4 Status of the low-state branch

The streamed operator, row count, and correctness fallbacks above remain useful complementary machinery. No production numerical profile is reported because the artifact does not regenerate a selected multidegree, success-probability bound, and state figure from one checker. A numerical frontier would require both that profile generator and a production border-basis or solve transcript.

14 Implementation, evidence, and reproducibility

The companion artifact separates theorem checks from experimental evidence. The primary selected-degree-free conclusions do not rely on sampled production ranks or small-profile Macaulay behavior. Formula checkers verify the numerical consequences of the stated hypotheses; they do not verify those hypotheses on an official key. Two separate correspondence tests exercise the reduction on one official key and the complete zero-offset wrapper at an unofficial toy shape. Neither executes the production-size nonlinear solver.

Table 4: Evidence shipped with the release. "Included" means that the named script and its pinned input are present, not that the headline recovery cost has been measured.

Evidence	Availability	What it establishes
Formula and ledger checks	Included	Numerical consequences of the dimensions, probability models, and cost conventions supplied to the scripts; not scheme correspondence or solver execution.
Scalar circuit netlist	Included	The emitted 150-gate $\mathbb{F}_{19}$ multiplication netlist is correct on all 361 valid inputs. The extension-field figures are recurrence counts, not emitted netlists.
Official KAT verification and reduction	Included for one Level-I $(28, 5, 4, 4)$ key	A Python transcription parses the compressed public key, accepts the supplied signature, rejects an altered message, and obtains quotient/affine ranks 50/30. The resulting 50-equation system in 102 variables is left unsolved.
Reduced zero-offset forgery	Included at unofficial $(2, 1, 2, 2)$	The complete restricted map is interpolated; its image equals the explicit rank-three quotient image, leaving one target-consistency coordinate before public slice search. The main case, eight committed fresh-key regressions, and a separate 24-forgery/100-rank stress check pass in the shared KAT-anchored Python transcription. This is toy composition evidence, not independent C-verifier or production-cost evidence.
All-shape official-key preflight census	Not included	No empirical pass probability for the fixed-layout or realized-rank hypotheses.
Production-size solver or forgery	Not included or claimed	No empirical evidence for H_{hom}, κ_{hom}, κ_{JG}, memory, or feasibility at a published shape.

14.1 Exact checks in the release

The release contains machine-readable ledgers and deterministic checkers. The main all-nine checker recomputes dimensions, separator choices, adaptive mixtures, target-sampling acceptance, and repair floors, while the separate field-circuit checker verifies the tower identities and all $19^2 = 361$ scalar products. The Just Guess checker recomputes the expected-operation Boolean ledger, the capped-tree sensitivity, the quadratic-root truth table, the cross-channel second moment, and every displayed break-even threshold. These are consistency checks, not independent evidence for their input assumptions.

The official-key harness uses only the compressed public key for the attack reduction. On the first published Level-I KAT it verifies the supplied signature, checks the substituted public map against a separate literal evaluator, and obtains the 50/30 quotient/affine ranks. It does not solve the residual 50-in-102 MQ instance or output a forgery. A second script uses the same Version 2.3 conventions at the deliberately unofficial shape $(v, o, \ell, r) = (2, 1, 2, 2)$. It imposes $X_i = [u_i \mid 0]$, interpolates the complete restricted verifier, checks that its output image equals the explicit rank-three symmetric quotient image, and leaves one target-consistency coordinate. It fixes the SHAKE target before enumerating public three-dimensional slices, reconstructs and serializes a signature, and checks the rejection rule. The main case and eight fresh-key regressions are recomputed; a separate executable test passes 24 further forgeries and obtains rank three on 100 further reduced keys. Positive and negative checks use a distinct literal evaluator in the shared KAT-anchored Python transcription. The forger and verifier paths still share scheme helpers, so this is a composition regression rather than independent verifier evidence. It does not call the unmodified official C verifier and it is not evidence about a production-size solver. The public leading-Jacobian preflight is deterministic on a fixed key; its failure invokes a complete-core fallback. The separate structural preflight is an exact condition for the displayed $\ell = 4$ routes and is not assigned a random success probability in the ledger.

14.2 What is not claimed

The release does not contain a production-size end-to-end forgery, a test of $\mathsf{H}_{\mathrm{hom}}$ on a published residual ideal, or a measured symbolic-homotopy implementation. In particular, the paper does not claim that κ_{hom} is small, that the direct parametrization fits in commodity memory, or that the unit-leading exponent is a complete NIST gate count. The AXN basis prices the multiplication schedule selected by the arithmetic theorem. Control, data movement, indexing, factorization, communication, and peak state remain outside that leading schedule and inside κ_{hom}. In the Just Guess branch, all field operations in the published expected schedule are converted to explicit Boolean-circuit costs and separate control/filtering reserves are displayed, but the candidate-generation regularity multiplier κ_{JG} remains to be measured or certified on a production key.

The low-state XL branch has the opposite profile: its streamed operator and state are explicit, but its production selected degree is not proved. A production border-basis transcript or a full-size solve would settle that remaining branch-specific question without changing the exact quotient or the selected-degree-free theorem.

14.3 Proposed production fixed-key record

The release does not supply the following production record. A deterministic record for one public key would consist of:

1. the public relation, output split, and all exact rank preflights;
2. the selected slice and either its exact projective rank histogram or a stronger unit-ideal certificate;
3. the fast-core leading-Jacobian witness, if the fast branch is taken;
4. the homotopy random coins, parametrization, and all candidate filters;
5. the reconstructed signature and the unmodified verifier transcript.

Such a record would replace the idealized transcript statement for that key. Only an instrumented implementation carrying all omitted work could measure κ_{hom} and resident state.

15 Necessary dimension floors within the common-column model

The common-column feature-count floor and the fresh-slice floor are independent necessary conditions for eliminating two dimension guarantees while retaining the stated architecture. Neither condition is sufficient for security or insecurity.

Proposition 15.1 (Dimensionally square common-column slice bound)**.** *Let $A = \mathbb{F}_{q^d}$ and retain the public vinegar summand V' of base-field dimension $d(v-1)$. Suppose an exact affine reduction has M' outputs, K' quotient quadratics, and full affine rank $M' - K'$. Then*

$$\dim_{\mathbb{F}_q}(V' \cap \ker C_{R,v}) \geq d(v-1) - (M' - K').$$

If $M' \leq d(v-1)$, dimensions alone leave an ordinary K'-dimensional slice carrying all K' residual equations. Equality of equation and variable counts does not establish a root, zero-dimensionality, nonsingularity, or an efficient solve. Eliminating this dimensional guarantee requires $M' > d(v-1)$. Under the coupling $M' = dx'$, this forces $x' \geq v$ and $M' \geq dv$.

Proof. Use
$$\dim(V' \cap \ker C_{R,v}) \geq \dim V' - \operatorname{rank} C_{R,v} = d(v-1) - (M' - K').$$
The right side is at least K' exactly when $M' \leq d(v-1)$. Negate and round to the next multiple of d. □

Let $K_{\mathrm{cc,ord}} = \min\{M, m_1 d^2\}$ be the pre-symmetrization ordered-label cap within the same common-column model. Retaining symmetric forms requires $m_1' \geq \lceil K_{\mathrm{cc,ord}} / \binom{d+1}{2} \rceil$ and $M' \geq K_{\mathrm{cc,ord}}$. Under $M' = dx'$ and $m_1' = \lceil x'/d \rceil$, all three necessities combine as follows.

Theorem 15.2 (Combined necessary common-column output floor)**.** *Any redesign that retains symmetric public forms, the public vinegar count, the public-vinegar affine geometry above, full affine rank $M' - K'$, and the Version 2.3 output/base-form coupling, while both restoring $K_{\mathrm{cc,ord}}$ and eliminating the dimensionally guaranteed fresh slice, must satisfy*

$$x' \geq \max\left\{ v, \left\lceil \frac{K_{\mathrm{cc,ord}}}{d} \right\rceil, d\left(\left\lceil \frac{K_{\mathrm{cc,ord}}}{\binom{d+1}{2}} \right\rceil - 1 \right) + 1 \right\}, \qquad M' = dx'. \tag{64}$$

For the nine official rows:

Level	*parameters*	*current M*	*necessary M'*	*increase*
I	$(28, 5, 4, 4)$	80	116	45.00%
I	$(48, 16, 2, 2)$	64	96	50.00%
I	$(28, 4, 4, 5)$	80	116	45.00%
III	$(40, 7, 4, 4)$	112	180	60.71%
III	$(72, 24, 2, 2)$	96	144	50.00%
III	$(38, 5, 4, 5)$	100	152	52.00%
V	$(50, 9, 4, 4)$	144	228	58.33%
V	$(96, 32, 2, 2)$	128	192	50.00%
V	$(52, 6, 4, 6)$	144	228	58.33%

Thus a parameter-only redesign that insists on both dimension goals while retaining these hypotheses needs at least 45.00% *more verifier outputs, and the largest necessary increase is* 60.71%. *These are necessary, not sufficient, dimensions; they do not prove any enlarged shape secure.*

Proof. The second and third terms in (64) are the two ordered-cap requirements under $m_1' = \lceil x'/d \rceil$. The term v is Theorem 15.1. Their maximum is therefore necessary. Substitution gives the table. □

15.1 A stronger zero-offset common-column dimension floor

The preceding floor assumes a full-rank affine lift and asks dimensions to remove a square slice from every target fiber. A zero-offset attacker has a second option: give up affine consistency for all targets, filter chosen-message targets through the quadratic image, and recover the entire common-column domain. Eliminating that dimensional configuration alone is much more expensive.

Proposition 15.3 (Zero-offset dimensionally square restriction)**.** *Consider any symmetric redesign for which the zero-offset common-column relation with $R_0 = 1$ is defined. Let Q_R be its quadratic output matrix and put $\rho_Q = \operatorname{rank} Q_R$. Public row reduction gives an invertible output change under which the zero-offset verifier is equivalent to*

$$g_{\rho_Q}(u) = z \in \mathbb{F}_q^{\rho_Q}, \qquad t_{\text{con}} = 0 \in \mathbb{F}_q^{M'-\rho_Q}, \tag{65}$$

where g_{ρ_Q} consists of ρ_Q independent quadratic output combinations. For an exactly uniform target, the consistency event in (65) has probability $q^{-(M'-\rho_Q)}$ and, conditioned on it, z is uniform in $\mathbb{F}_q^{\rho_Q}$.

With zero offsets no reserved line is needed, so use the full public-vinegar space $V_{\text{vin}} \cong A^v$, of base-field dimension dv. If

$$\rho_Q \leq dv, \tag{66}$$

then dimensions alone leave an ordinary ρ_Q-dimensional slice $L \subseteq V_{\text{vin}}$ carrying all ρ_Q residual equations: it has ρ_Q equations in ρ_Q variables. This dimension equality alone proves neither root existence nor efficient recovery. Consequently, ruling out this chosen-message dimensional configuration requires

$$\rho_Q > dv. \tag{67}$$

Proof. With zero offsets, the substituted verifier is $Q_R z_{\text{sym}}(u) = t$ and contains no affine or constant term. Choose a row basis of Q_R and extend it to an invertible basis change on the M' output coordinates. The first ρ_Q transformed rows are the independent residual quadratics g_{ρ_Q}; the remaining rows vanish identically on the quadratic image and therefore impose the consistency equations in (65). An invertible linear transformation sends a uniform target to independent uniform residual and consistency coordinates, proving the distribution statement. Under (66), choose any ρ_Q-dimensional linear subspace of V_{vin} and restrict all ρ_Q residual equations to an affine coset of it. This gives the asserted dimensionally square restriction. $\square$

Symmetry bounds the actual quotient rank by

$$\rho_Q \leq \min\left\{M',\ m_1'\binom{d+1}{2}\right\}. \tag{68}$$

Hence making (67) even possible forces both terms on the right of (68) above dv.

Theorem 15.4 (Output floor eliminating the zero-offset dimension guarantee)**.** *Retain symmetry, the public vinegar count, and the Version 2.3 coupling*

$$M' = dx', \qquad m_1' = \left\lceil \frac{x'}{d} \right\rceil.$$

Put

$$C_d = \binom{d+1}{2}, \qquad m_{\text{sq}} = \left\lfloor \frac{dv}{C_d} \right\rfloor + 1.$$

A necessary condition for eliminating the zero-offset chosen-message dimension guarantee is

$$x' \geq \max\{v+1,\ d(m_{\mathrm{sq}}-1)+1\}, \qquad M' = dx'. \tag{69}$$

If the redesign also restores the common-column ordered-label cap $K_{\mathrm{cc,ord}}$*, the right side of* (69) *must additionally be maximized with*

$$\left\lceil \frac{K_{\mathrm{cc,ord}}}{d} \right\rceil, \qquad d\left(\left\lceil \frac{K_{\mathrm{cc,ord}}}{C_d} \right\rceil - 1\right) + 1.$$

For the nine official rows, the zero-offset condition already dominates the ordered-cap terms:

Level	*parameters*	*current M*	*dimension-only M′*	*increase*
I	$(28,5,4,4)$	80	180	125.00%
I	$(48,16,2,2)$	64	130	103.125%
I	$(28,4,4,5)$	80	180	125.00%
III	$(40,7,4,4)$	112	260	132.14%
III	$(72,24,2,2)$	96	194	102.08%
III	$(38,5,4,5)$	100	244	144.00%
V	$(50,9,4,4)$	144	324	125.00%
V	$(96,32,2,2)$	128	258	101.5625%
V	$(52,6,4,6)$	144	324	125.00%

Thus eliminating this dimension guarantee under the stated coupling requires 101.5625%–144.00% *more outputs across the published rows.*

Proof. Condition (67) and the two caps in (68) require $dx' > dv$ and $m_1' C_d > dv$. The first inequality is equivalent to $x' \geq v+1$. The second requires $m_1' \geq m_{\mathrm{sq}}$, and the least integer x' with $\lceil x'/d \rceil \geq m_{\mathrm{sq}}$ is $d(m_{\mathrm{sq}}-1)+1$. This proves (69). The two extra ordered-cap terms are the same feature and output necessities used in Theorem 15.2. Direct substitution gives the table. □

How to interpret the stronger floor. This is deliberately a dimension-only theorem, not a claim that every secure repair must literally double its outputs. A redesign below the table can instead leave the square system present and try to obtain security from the chosen-message consistency factor $q^{M'-\rho_Q}$, from the cost of solving the square system, or from a changed target interface. It must then publish and price those ingredients explicitly. The displayed floor applies only to a redesign that insists on removing this dimension guarantee while preserving the stated common-column architecture and parameter coupling.

16 Conclusion

After the public common-column relation, the $q = 19$ Version 2.3 construction loses quadratic directions for every symmetric public key. Ordered power-pair labels are not ordered scalar polynomials in the remaining column: swapping the labels leaves that restricted quadratic unchanged before the public outer map. This does not assert the same cap for the unrestricted matrix-valued verifier, where label exchange also transposes the column indices. Conditional on the exact public ranks, output splitting turns the restricted identity into square residual systems without discarding a verifier equation, and the block-symmetry rejection rule does not remove them.

Every published parameter shape has the required dimensions; application to a supplied key remains conditional on the stated structural and spectrum preflights. The finite-field homotopy adaptation gives a selected-degree-free, fixed-core-free recovery algorithm only under those hypotheses and $\mathsf{H}_{\mathrm{hom}}$, but its all-nine numerical comparison is a unit-leading break-even ledger with an explicit, unmeasured κ_{hom}. The sufficient adaptive thresholds, rounded downward, are 10.811927, 23.205374, and 38.156013 bits of allowable overhead at Levels I, III, and V; the paper does not identify those values with complete classical-gate bounds.

The Just Guess route gives a more explicit sensitivity for the Level-I $\ell = 2$ shape. It replaces symbolic homotopy by a Boolean-ledger instantiation of the published expected schedule of guesses, quadratic roots, linear solves, quotient checks, and verifier calls. The expected trial ledger is below $2^{132.047}$; an exact conditional-channel atom calculation and a cross-channel second moment give a success-adjusted exponent below $135.425325 + \log_2 \kappa_{\mathrm{JG}}$. Thus the Level-I comparison is below 2^{143} under the sufficient condition $\log_2 \kappa_{\mathrm{JG}} < 7.574675$. The candidate event is transcript-checkable, but its multiplier is unmeasured and can be infinite. This is therefore still a premise, not an unconditional complete-gate break.

The release is therefore evidence for an exact structural reduction and conditional attack architecture, not an empirical demonstration of a production-size forgery. Its machine checks establish only the statements listed as available in Table 4; they do not by themselves establish the official EUF-CMA security experiment.

A redesign that insists on both restoring the pre-symmetrization common-column label cap and eliminating the fresh-slice dimension guarantee, while retaining the current coupling, requires 45.00%–60.71% more outputs. Eliminating the zero-offset common-column dimension guarantee under the same coupling requires 101.5625%–144.00% more. These are necessary dimension floors for those specific goals, not general lower bounds on every secure repair.

A A finite fixed-key spectrum-certificate format

The idealized XOF-transcript theorem is not required for a statement about a concrete key. This appendix gives a finite exact certificate format for the aggregate spectrum. No such production-key spectrum certificate is shipped in the release.

Put $N_U = \dim_{\mathbb{F}_q} U$. Choose a basis $y_0, \ldots, y_{h-1}$ of a fixed slice L and write

$$T(Y) = \sum_{i=0}^{h-1} Y_i T_{y_i} \in \mathbb{F}_q[Y_0, \ldots, Y_{h-1}]^{K \times N_U}.$$

For $0 \le j < h$, use the disjoint normalized projective chart $Y_0 = \cdots = Y_{j-1} = 0$, $Y_j = 1$, and put $R_j = \mathbb{F}_q[Y_{j+1}, \ldots, Y_{h-1}]$. Let $T^{(j)}$ be the resulting matrix. For $t \ge 0$, define

$$I_{j,t} = \left\langle Y_i^q - Y_i \ (i > j), \quad \text{all } (t+1) \times (t+1) \text{ minors of } T^{(j)} \right\rangle \subseteq R_j. \tag{70}$$

Put $n_{j,\le t} = \dim_{\mathbb{F}_q}(R_j / I_{j,t})$, $n_{j,\le -1} = 0$, and

$$N_t = \sum_{j=0}^{h-1} (n_{j,\le t} - n_{j,\le t-1}). \tag{71}$$

Proposition A.1 (Projective rank-histogram certificate)**.** *The integer N_t is exactly the number of projective directions $[y] \in \mathbb{P}(L)(\mathbb{F}_q)$ for which* $\operatorname{rank} T_y = t$*. Consequently*

$$\bar{\eta}_L^U = (q-1) \sum_t N_t q^{-t}. \tag{72}$$

Reduced Gröbner bases with their standard monomial sets, or polynomial reduction transcripts proving the unit-ideal and quotient-dimension claims, form machine-checkable fixed-key certificates.

Proof. The normalized charts contain exactly one representative of every projective $\mathbb{F}_q$-direction. The field equations identify

$$R_j/\langle Y_i^q - Y_i : i > j\rangle \cong \prod_{a\in\mathbb{F}_q^{h-j-1}} \mathbb{F}_q$$

by evaluation. Every ideal in this product of fields is radical, and its quotient dimension is the number of coordinates on which all generators vanish. The minors in (70) vanish exactly when $\operatorname{rank} T^{(j)}(a) \le t$. Thus $n_{j,\le t}$ counts rank-at-most-t points in chart j, and differencing gives (71). Every projective direction has $q-1$ nonzero scalar representatives and rank is invariant under base-field scaling, proving (72). □

A stronger certificate that $\operatorname{rank} T_y \ge t_0 + 1$ for every nonzero y consists of unit-ideal transcripts for all I_{j,t_0}. The complete histogram is more flexible: a small exceptional low-rank locus can be charged exactly instead of controlling the whole pencil by its minimum rank.

B Dimension formulas and exact numerical rows

For a Version 2.3 row,

$$n = v + o, \quad m_1 = \left\lceil \frac{or}{\ell} \right\rceil, \quad M = or\ell, \quad N_0 = n\ell, \quad K = m_1 \binom{\ell+1}{2}.$$

For a full affine lift with $\operatorname{rank} C_{R,v} = M - K$,

$$\dim\ker C_{R,v} = N_0 - M + K.$$

For a structured lift whose complete offset orbit has rank $m_1\ell^2$,

$$\dim_A H^F_{R,v} = n - m_1\ell.$$

The direct ordinary fresh-slice lower bound for $\ell = 4$ is

$$\dim(V' \cap \ker C_{R,v}) \ge 4(v-1) - (M-K).$$

Table 5: Exact unit-leading exponents underlying the rounded main text. Every row conditions on its exact public quotient/rank preflights, the applicable spectrum/model events, and $\mathsf{H}_{\mathrm{hom}}$; the $\ell = 4$ rows additionally condition on the fixed-layout structural preflight. The values exclude $\log_2 \kappa_{\mathrm{hom}}$. "Direct" solves the complete square; "adaptive" is the conditional mixture of an exact fast-core preflight and the complete-square fallback.

Level	Parameters	Direct	Adaptive	Adaptive headroom
I	$(28,5,4,4)$	142.606960	128.246278	14.753722
I	$(48,16,2,2)$	132.970670	132.188072	10.811928
I	$(28,4,4,5)$	142.606374	128.245747	14.754253
III	$(40,7,4,4)$	185.450879	171.616678	35.383322
III	$(72,24,2,2)$	183.794625	183.794625	23.205375
III	$(38,5,4,5)$	185.450045	171.615857	35.384143
V	$(50,9,4,4)$	227.560406	214.558321	57.441679
V	$(96,32,2,2)$	233.843987	233.843987	38.156013
V	$(52,6,4,6)$	227.559359	214.557277	57.442723

All values charge the 150, 692, and 2628-gate multiplication circuits as appropriate and use the conservative B^2 forbidden-vector separator. For $\ell = 4$, the Jacobian-preflight density uses the projective right-kernel union bound, not an independence claim after a biased linear transform.

C Arithmetic-circuit certificate details

The scalar multiplier consumes two little-endian five-bit canonical integers in $\{0, \dots, 18\}$ and returns the canonical five-bit representation of their product modulo 19. Its 150 gates are two-input AND, XOR, and XNOR gates with free constants, wires, fan-out, and structural sharing. The released netlist is tested on all 361 valid input pairs. The SHA-256 digest of the canonical compact JSON payload containing exactly `ninputs`, `gates`, and `outputs` is

`e1708a32c1475ad14295c2b34fed976fb09577b127810d29bf237db18aabc131`

for the emitted balanced-netlist representation. This is an internal payload identifier, not the bytewise digest of the surrounding ledger file.

The tower recurrences are

$$\mathbb{F}_{19^2} = \mathbb{F}_{19}[u]/(u^2+1), \qquad \mathbb{F}_{19^4} = \mathbb{F}_{19^2}[v]/(v^2-(1+u)).$$

Three-product Karatsuba at each level, together with the exact linear operations stated in Theorem 7.1, gives constructive recurrence counts 692 and 2628. The artifact verifies the tower identities, the basis determinant, and the recurrence arithmetic; it does not emit or exhaustively test complete extension-field netlists.

These circuits normalize only the pointwise multiplication schedule. They are not lower bounds and do not price the whole solver. Additions, inversions, factorization, bulk evaluation/interpolation transforms, filtering, control, indexing, communication, and memory remain in κ_{hom}.

D Reproduction and package contents

The source package is self-contained. Its root contains the complete paper source, bibliography, compiled PDF, build instructions, and hashes. The `artifact/` directory contains:

- a standalone generator and separately coded consistency checker for the all-nine conditional homotopy ledger;
- the complete 150-gate scalar multiplier netlist and tower checks;
- the Just Guess expected-operation and capped ledgers, the quadratic-root check, and the exact conditional Frobenius-channel atom enumeration;
- a public-key correspondence harness for one official Level-I KAT; and
- a non-planted zero-offset end-to-end composition test at the unofficial shape $(2, 1, 2, 2)$, with fresh-key forgery and rank stress checks.

A clean reproduction is:

```
sha256sum -c SHA256SUMS
make regenerate
make verify
make
```

The all-nine generator assigns no random probability to the fixed outer map: its $\ell = 4$ ledgers condition on the exact public structural preflight. The checkers are deterministic, require Python 3.10 or newer plus NumPy for the two correspondence tests, and do not require network access. The paper build uses only the files in the package. The supplied `SHA256SUMS` records every distributed source and pinned input; generated TeX build files are excluded.

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